ANALYSIS

The data economy of public proteomics: thirteen years of PRIDE

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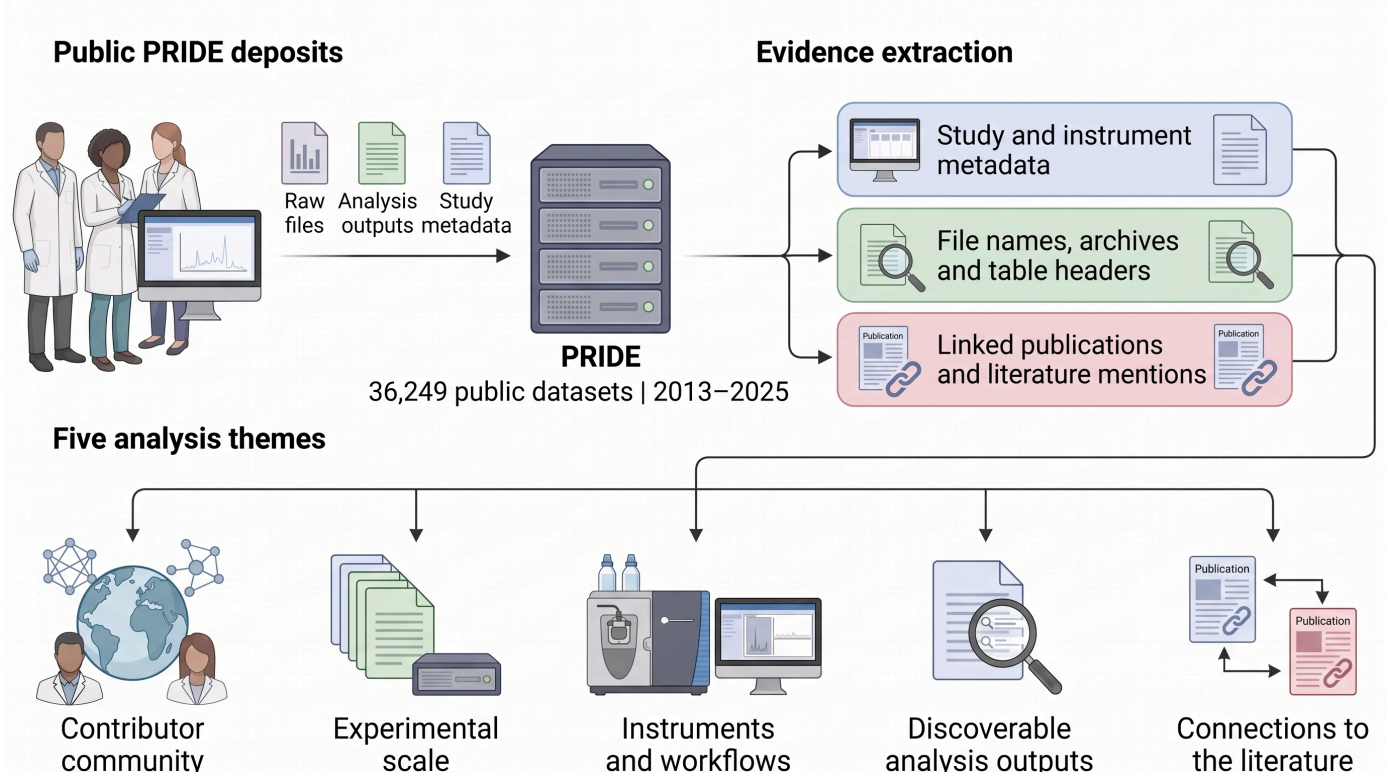
Pre-submission review · October 2026

PRIDE's public archive records how proteomics research changes as its community grows. We examined all 36,249 datasets released between 2013 and 2025 using study records, deposited files and links to published papers. The contributor community broadened, and linked papers appeared in a wider range of journals. Large datasets became more numerous, although the median number of mass-spectrometry runs per dataset fell. New instruments and software appeared in deposits alongside older ones. We recognised peptide or protein identification results in 80% of 2025 datasets by reading files produced by individual analysis programs; 16% contained results in community-standard formats. We detected no association between those standard formats and dataset mentions in the other papers we analysed. Mentions alone cannot show that the data were analysed again. Searching the formats used by individual analysis programs could help readers find more of the results already shared through PRIDE.



Created with BioRender.

Introduction

Researchers prepare proteomics data and results for public release and publication. Later work can build on them. We use data economy to mean this circulation of data after an experiment. Costs and prices are outside our scope. PRIDE, the largest repository in the ProteomeXchange network of proteomics data repositories, preserves mass-spectrometry proteomics data from a growing community, makes them publicly accessible and links deposits to the published literature[1,2].

Evidence in PRIDE's deposited files connects the growth recorded in repository reports to changing experiments and analytical workflows. Raw files record experimental runs and instrument labels. We identify the analysis software from logs and parameter files. With dates and publication links, these records trace the history of a study and make changes in research practice visible across the archive.

An archive-wide view of PRIDE complements studies of public-data impact through citations[3] and examinations of selected deposits. These examinations include six reanalysed projects[4] and an audit of 500 datasets[5]. Deposited files also show how developments described elsewhere, including large cohorts[6,7] and standard result formats[8], appear in the public record.

We examine all 36,249 PRIDE datasets released between 2013 and 2025. We follow the expanding contributor community by comparing dataset sizes and recorded workflows, then examine result files to identify analytical content beyond standard result formats. Literature links extend this account to publication and later attention. The analysis describes deposits rather than instrument ownership or adoption dates. For each analysis, we report the number of records used as its denominator and, where computed, 95% intervals from resampling contributor groups. We approximate these groups

by linking datasets through identifiers derived from principal-investigator contacts. Mentions of PRIDE dataset identifiers (accessions) in later papers provide an incomplete proxy for reuse and do not establish reanalysis.

Results

PRIDE reached a broader contributor community

PRIDE expanded across contributor groups and the journals publishing their work. Comparing 2023 to 2025 with 2013 to 2015, the effective number of groups increased 5.0-fold and that of journals 5.4-fold (95% interval 4.7 to 6.3) (Fig. 1a,b; Supplementary Tables S2 and S3). By publication year, the number of distinct journals publishing linked papers rose from 78 in 2013 to 627 in 2025 (Supplementary Table S18). Effective numbers account for both the number of categories and how evenly datasets are distributed among them (Box 1).

Broader participation accompanied more even representation of instrument models. Their effective number increased 1.8-fold (1.6 to 2.0), although the number of distinct models expected in equal-sized samples did not increase (61 against 66 per 1,500 datasets). The effective number of species decreased to 0.67 times its earlier value (0.58 to 0.81). For contributor groups, journals, instrument models and species, the direction of change agreed between orders 1 and 2, which give different weights to common categories (Supplementary Table S3).

At order 2, Supplementary Table S3 shows no detectable change in the effective numbers of analysis tools, research subfields, countries and instrument vendors. The effective numbers increased at different times: instrument models mainly from 2020, journals until 2022, and tools only in 2024 and 2025 (Fig. 1a).

Human and mouse accounted for most of the concentration in species. Human samples were named in 41% of datasets released in 2013 and 52% of those released in 2025; mouse was named in 11% and 19%. Datasets naming neither species made up 46% of deposits during 2013 to 2016, compared with 33% during 2023 to 2025 (Supplementary Fig. 9). Among the 354 groups contributing in both periods, this share also declined (from 40% to 32%), so changes in which groups contributed explained only part of the decline.

New contributors continued to supply a substantial share of PRIDE's deposits. Groups with no earlier PRIDE release since 2013 contributed between 42% and 48% of annual deposits during 2020 to 2025 (44% in 2025; Fig. 1c,d). Two-thirds of all groups deposited a single dataset, accounting for a quarter of all datasets (Tables S4 and S5; Supplementary Fig. 1).

The number of datasets from groups whose country could be identified increased 3.3-fold when comparing 2023 to 2025 with 2016 to 2018. Half of the additional datasets came from Europe (Fig. 1e to g; Supplementary Figs 10 and 11; Supplementary Table S6). Datasets from China increased 5.2-fold over these periods, compared with 2.8-fold for Germany. China's share of datasets whose country could be identified rose from 3% in 2013 to 15% in 2025. These counts concern PRIDE only and exclude groups that deposit only in other repositories, including iProX[9], jPOST[10] or MassIVE[2].

Archive growth encompassed experiments at different scales

PRIDE's growth[1] accommodated experiments at different scales. The median number of raw runs (mass-spectrometry acquisitions) per dataset was 16 for 2023 to 2025 (n = 15,248), compared with 30 for 2013 to 2016 (n = 2,498), a ratio of 0.53 (95% interval 0.48 to 0.63). The whole distribution shifted towards fewer runs (Fig. 2a).

Large deposits nevertheless became more numerous. The mean annual number of datasets with at least 100 runs was 124 during 2013 to 2016 and 459 during 2023 to 2025, while their share of all datasets fell from 19.8% to 9.0% (Fig. 2b). Such deposits are relevant to the scale proposed for proteomics studies of participant cohorts[6,7]. Run counts differ from sample counts because isobaric labelling uses tags to measure several samples together in one run. The mean number of channels per labelled dataset increased from 6 in 2014 to 13 in 2025 as TMTpro[11] replaced iTRAQ[12]; these labelling methods use channels to distinguish the samples (Supplementary Fig. 8).

The decrease in median runs persisted across five sensitivity analyses: combining datasets from the same paper; counting exact runs only; collapsing fraction-numbered runs; reweighting to the 2013 to 2016 composition of vendor, acquisition mode, labelling, species and submission type; and stratifying by country (Fig. 2c; Supplementary Table S8). Datasets labelled with tandem mass tags (TMT)[13], and those naming SCIEX or Bruker instruments, showed no detectable decrease in median raw runs (early Bruker n = 56).

Among the 3,456 groups contributing in multiple years, the slope over time of the base-2 logarithm of raw-run counts, estimated within groups, was about 38% of the pooled slope on the same 23,486 datasets. For 2018 to 2025 releases, the matched ratio of geometric means of run counts in first-time versus returning groups was 0.64 (0.61 to 0.68). Geometric means average counts on a logarithmic scale; the comparison stratified datasets by release year, vendor, acquisition mode, human-sample status and continent.

Individual runs occupied more storage. Median raw-run size increased from 0.35 GB in 2013 to 1.07 GB in 2025. Median raw-run sizes were 0.44 GB on the LTQ Orbitrap[14] and 3.95 GB on the Orbitrap Astral[15] (Fig. 2d; Supplementary Table S7). Among datasets naming one instrument model and having exact run counts, we separated the change in mean log10 (base-10 logarithm) bytes per run between the 2014 to 2016 and 2023 to 2025 periods into changes in the mix of models and changes within each model. The component from the changing mix of models was 2.30-fold, and the component from changes within models was 0.85-fold (0.80 to 1.10). With fewer but larger runs, median dataset size remained between 19 and 25 GB in every year, while the mean and the sizes of the largest datasets grew.

Storage was concentrated in a small fraction of deposits. The largest 1% of datasets held 24% of the 2.71 petabytes (PB), and the largest 10% held 61% (Fig. 2e). Raw files in instrument manufacturers' formats accounted for 84% of all bytes (Fig. 2f; Supplementary Fig. 6).

Deposits recorded technological change alongside established workflows

PRIDE's instrument labels document changing workflows alongside persistent earlier technologies. For these counts, each dataset with an identified instrument model contributed one count, divided equally among its named models. Across years, Thermo Fisher labels accounted for 79 to 89% of these fractional counts; the repository reports a similar share of stored raw files[1]. The LTQ Orbitrap accounted for 40% of these fractional counts in 2013, the first year of the series. Successive leading models had progressively lower peak shares of these fractional counts, with 38% for the Q Exactive[16] and 19% for the Q Exactive HF[17] (Fig. 3a; Supplementary Fig. 2).

After combining labels that refer to the same instrument model, the effective number of models rose from 4.7 (95% interval 4.0 to 5.7) in 2013 to 12.1 (11.5 to 12.8) in 2025 (Fig. 3b). Every model named in at least 50 datasets was still represented among 2025 deposits.

This record includes reanalyses and data released after a delay from the experiment. Datasets labelled with timsTOF models[18] represented 13% of the 2025 total, compared with 1.8% in 2021. This instrument family had been described in 2021 as becoming widely used[19].

New models appeared in deposits from Asia about one year later than expected from its dataset count. Africa's longer delay was close to that expected from its small number of datasets (Supplementary Fig. 3). These expectations came from drawing instrument-model labels independently of continent within each release year, without accounting for similarities among datasets from the same contributor group.

DIA-NN[20] was named in a rapidly growing number of deposited datasets. It was named in 2 datasets released in 2020 and 722 in 2025, ranking third behind MaxQuant[21] (28% of datasets) and Proteome Discoverer[22] (20%) (Fig. 3c).

Tool use also varied by country. In 18 of 20 pairs of a tool and its developers' country, groups in the developers' country named the tool more often than groups in all countries combined. The median ratio across pairs was 1.8-fold (permutation P = 0.0002; Fig. 3d). For 11 pairs, the 95% interval excluded a ratio of 1, which would indicate equal naming rates. Examples include Spectronaut[23] in Switzerland (3.4-fold), Comet[24] in the United States (2.9-fold) and MaxQuant in Germany (1.5-fold). The developers' own laboratories were not excluded. A similar pattern had been reported for open-source analysis workflows in western Europe[22].

Analysis files preserved the software versions used. Parameter files and logs identified MaxQuant versions in the 1.6 series, first recorded in PRIDE in 2017. These versions accounted for 47% of MaxQuant datasets released in 2025 for which a version could be recovered. Among 2025 submissions with a recoverable version, the share was 40%; versions were recoverable for 361 of 576 datasets naming MaxQuant. The DIA-NN 1.8 version series accounted for 61% (48 to 74%) of DIA-NN datasets submitted in 2025 with a recoverable version. Versions were recoverable for 110 of 331 submissions naming DIA-NN (Supplementary Fig. 4 and Table S12).

Older measurements were associated with older software versions. We compared MaxQuant datasets submitted in 2023 to 2025 by the age of their newest raw file at submission. MaxQuant versions beginning with 1 were used in 35% of datasets when that file was less than one year old and in 80% when it was more than three years old.

Tool-specific files revealed analytical content beyond standard formats

The deposited files contained far more recognisable peptide or protein identification results than standard formats alone revealed. Among datasets released in 2025 (n = 6,231), reading table column headings brought detection of identification results to 80%, compared with 16% containing standard result files. Detection had reached 59% from file names and 68% after inspecting zip contents (Fig. 4c; Supplementary Table S13). Even after searching file names, zip contents and table column headings, the detectable share fell from 92% in 2015 to 80% in 2025 (Fig. 4d). These searches can miss files that contain identification results, so the detected shares are lower bounds. Detection alone cannot show whether a file is valid or usable.

Proteomics Standards Initiative formats provide common representations of results; mzIdentML has been implemented in over 35 software tools[8]. We counted submitter-deposited mzIdentML[25] or mzTab[26] as standard result files and excluded repository-generated files (Box 1). Standard result files appeared in 16 to 20% of datasets each year since 2015 (Supplementary Table S13).

The share of datasets containing standard result files differed across the named analysis tools. Among datasets naming a single analysis tool, standard result files appeared in 0.2% for DIA-NN and 76% for MS-GF+[27], the lowest and highest rates. Spectronaut and MaxQuant also had low rates; Mascot[28] deposits more often included standard files (Fig. 4a). Supplementary Table S9 gives the tool-specific rates.

Among datasets naming tools that write standard files by default, 55% contained these files (95% interval 44 to 66%). The rate was 15% (14 to 18%) for datasets naming tools that offer standard files as an option and 3% (2 to 4%) for datasets naming tools that cannot write them (Fig. 4b; Supplementary Table S10). Only MS-GF+ and MetaMorpheus[29] made up the default-writing group (144 datasets). MetaMorpheus deposits included standard results in 23% of cases despite the software writing them by default.

The mix of analysis tools named in datasets also changed over time. Between the 2019 to 2021 and 2023 to 2025 periods, the archive-wide standard-file rate changed by −1.4 percentage points (−3.0 to 0.3). Separating changes in the mix of software families from changes within families assigned −3.0 percentage points (−3.4 to −2.5) to the changing mix, mainly the growth of DIA-NN and Spectronaut and the decline of Mascot. Changes in standard-file rates within software families contributed +1.6 percentage points (0.1 to 3.2).

Data-independent acquisition (DIA) fragments precursor ions within predefined mass-to-charge windows[30]. Among datasets with no DIA detected, the standard-file rate remained at 19 to 21% each year since 2020. The share of datasets with detected DIA grew from 2.8% in 2015 to 21.1% in 2025, but about 3% of DIA datasets released in 2023 to 2025 included standard result files. Differences in the formats used for spectral libraries, collections of reference spectra, have been described as one obstacle to DIA reanalysis[31].

Files containing information for individual mass-spectrometry scans, together with intermediate search outputs, occupied 66% of the bytes in uncompressed results archives, compared with 3.9% for final result tables (Supplementary Fig. 6). Files storing mass spectra in open formats appeared in 34% of datasets in 2013 to 2016 and 21% in 2023 to 2025. The halving of submissions classified as complete, a category requiring lists of measured spectral peaks, accounted for 5 to 10 of the 13 percentage-point decline (Supplementary Fig. 7).

Documentation was uncommon but often recurred within contributor groups. When a group's previous dataset included a README documentation file, its next did so in 41% of cases, against an expected 1.5% for datasets released in the same years. For sample and data relationship format (SDRF) files[32], which link sample descriptions to data files, the corresponding rates were 44% and 2.5% (Fig. 4e; Supplementary Table S14). These recurrence rates are unadjusted for pairs submitted on the same day or for the same paper. Some SDRF files were added to several datasets of a group after release. Across the archive, SDRF presence rose from 0.7% in 2020 to 2022 releases to 2.7% in 2023 to 2025, consistent with reports of limited use[33].

Release dates placed public deposits in the history of an experiment

Dates preserved in deposited files connected experiments and analyses to public release (Fig. 5a; Supplementary Table S15). Raw-file names supplied a measurement date for 11,868 of the 35,364

datasets with raw files (34%). In this subset, the median time from the latest measurement to public release was 2.2 years (95% interval 2.1 to 2.3).

Deposited files also provided dates for resources used in analysis. The date in the main sequence database file name (n = 2,558) preceded public release by a median of 2.5 years (2.4 to 2.6). The median age of the software release used (n = 10,466), measured from its first appearance in PRIDE, was at least 3.4 years (3.3 to 3.5; Supplementary Fig. 5 and Table S11).

Of the 1,006 datasets with both dates, 71% had a database file dated earlier than the latest measurement. The software release had appeared in PRIDE before the latest measurement in 79% of 3,986 datasets.

Most of the interval from measurement to public release elapsed before submission (Supplementary Table S15). Submission preceded publication of the linked paper by a median of 141 days; the median delay between publication and public release was 25 days.

Public releases also clustered in time. From October 2018 onwards, at least 50 datasets were made public on each of 62 days. Datasets released on these days became public a median of 95 days after publication of their paper, compared with 11 days on other days (Supplementary Fig. 13 and Table S16). Who set each release date is unrecorded. Supplementary Fig. 14 shows the separate record of submission timestamps.

PRIDE-linked publications reached beyond specialist journals

PRIDE-linked papers reached an increasingly broad range of journals. Journals with titles containing "proteom" published 51% of PRIDE-linked papers in 2013 and 10% in 2025, even as the number of linked papers they published rose from 142 to 259 (Fig. 5b,c; Supplementary Tables S17 and S18). The ten leading journals' combined share fell from 66% to 31% over the same period. In 2025, *Nature Communications* published the most PRIDE-linked papers of any single journal (Supplementary Table S17).

The research fields of linked papers also shifted. The share of linked datasets assigned to chemistry fell from 26% in 2013 to 6% in 2025. Almost all of this category was spectroscopy, the subfield to which mass-spectrometry methods papers are assigned. The share assigned to medicine rose from 18% to 29% (Supplementary Fig. 12). Coverage limits these comparisons: paper links were available for 94% of datasets released in 2013 but 61 to 67% of those released since 2021.

Accession mentions provided evidence of continued attention to public data

Text mining in Europe PMC[34] identified papers that named PRIDE dataset identifiers, called accessions. We included a mentioning paper if it was not marked as the dataset's primary reference, named datasets from at least two contributor groups and mentioned no more than 100 datasets. Papers meeting these criteria mentioned 2.3% of the 20,684 datasets released in 2013 to 2022 within the release calendar year or the next three calendar years (Fig. 5d; Supplementary Table S19). Within a calendar year, we could not determine whether the paper appeared before or after public release of the dataset. This measure gives an incomplete view of reuse, especially when accessions appear only in supplementary tables or literature outside open-access full text. A mention does not establish reanalysis.

Papers meeting these criteria named DIA datasets and datasets with human samples more frequently. The adjusted odds of a mention were higher for DIA datasets (odds ratio 1.71; 95% interval 1.22 to 2.35) and for datasets with human samples (1.39; 1.17 to 1.71). For datasets assigned only to Thermo Fisher, the estimated odds ratio was also above one, with an interval close to one, the value indicating no association (Fig. 5e; Supplementary Table S20). The model accounts for the other dataset and study characteristics specified in Methods; its DIA cohort includes releases only up to 2022.

We did not detect a relationship between standard result files and mentions in papers meeting our criteria (adjusted odds ratio 0.89; 0.66 to 1.17). The unadjusted mention-rate ratio for datasets with versus without standard result files was 1.04 (0.72 to 1.39) when the window covered the release calendar year and the next five calendar years, using datasets released by 2020. A restricted cohort of human datasets assigned solely to Thermo Fisher, with no DIA detected and naming MaxQuant or Proteome Discoverer, gave an imprecise unadjusted rate ratio of 0.74 (0.26 to 1.38) (Supplementary Table S20). A modest advantage in mention rates for datasets with standard result files therefore cannot be excluded.

SDRF files require a different interpretation. Datasets with SDRF files had higher odds of a mention in a paper meeting our criteria (odds ratio 4.08; 2.29 to 7.26), but 91 of these 205 datasets were released before the format existed. Those files were added later, often by reanalysis projects, so the direction of the association cannot be determined. Methods papers, machine-learning papers and papers presenting research resources were among those naming the largest numbers of datasets.

Discussion

Thirteen years of PRIDE deposits document the work of an expanding proteomics community. New contributors continued to account for a substantial share of annual releases, and linked papers appeared across a much wider range of journals. This collective record, built by the repository and its contributors, makes changes in research practice visible at a scale difficult to achieve through individual studies.

Examining deposited files shows which experiments contributed to PRIDE's growth. PRIDE accommodates increasing numbers of large deposits alongside many datasets with fewer raw runs. The mix of instrument models helps explain why a typical run occupies more storage. Older models remained represented in deposits, which include delayed releases and reanalyses. Deposited software versions also preserve evidence of earlier workflows. Participation can broaden even as biological coverage becomes more concentrated on human and mouse.

Identification results could be recognised in 80% of 2025 deposits when files from individual analysis programs were examined, although 16% contained standard result files. Recognising these outputs more systematically could make contributors' results easier to discover. File presence does not establish validity or readiness for reuse, so extraction would need validation and clear records of how the files were produced and processed.

Differences in deposition rates across analysis programs suggest where improvements could be tested with tool developers. Standard result files accompanied 55% of datasets whose tools write them by default; for tools lacking this capability, the rate was 3%. The low rate of standard result files for MetaMorpheus, which writes these files by default, shows that a default alone is insufficient, and tool choice may reflect depositor intentions. Sample annotations supplied during analysis could be used to create SDRF files, which link samples to data files, and reduce reconstruction work for later users[32,33]. Recording the instrument, software version and database date from

deposited files would help readers trace how the data were produced, a concern addressed in reporting recommendations for single-cell proteomics[35].

Release timing helps explain which stages of a study PRIDE records. Most of the measured time from data acquisition to public release elapsed before submission, so it cannot be attributed wholly to repository handling. Older software often accompanied older measurements. A public release date therefore needs to be read alongside the available acquisition and analysis dates.

Mentions of dataset identifiers in qualifying papers other than the dataset's primary reference provide an incomplete view of attention to deposits. These papers more often named DIA and human datasets. Standard result files showed no detectable association with those mentions, but the intervals do not exclude a modest benefit. Mentions alone do not establish reanalysis. Calendar-year mention windows cannot show whether a paper appeared before or after a dataset's release within the same year. The SDRF association also requires caution because many files were added after release, sometimes during reanalysis. Evaluations of future improvements should record when changes occur and distinguish subsequent use from metadata added during that use.

The scope and coverage of the evidence constrain these conclusions. PRIDE represents only part of ProteomeXchange, which matters especially for geographical comparisons. Identifiers derived from principal-investigator contacts approximate contributor groups: changed addresses can create apparent new groups, while core facilities serve several laboratories. Because the record starts in 2013, a group classified as first-time may have contributed before that year. File and text searches provide lower bounds on the presence of software, acquisition modes, labelling and identifiable results. Paper links cover 75% of datasets overall but 61 to 67% of recent releases; dated raw-file names cover 34% of deposits containing raw files. Text mining also misses later mentions, particularly those confined to supplementary material. The reconstructed snapshot cannot recover the history of metadata edits or withdrawn datasets and excludes recent submissions that remain private. Our comparisons are descriptive; the Supplementary Tables give denominators and intervals where computed, and Table S21 records the sensitivity analyses.

Making the analytical results already shared by PRIDE's contributors easier to find and documenting how those results were produced would extend the value of that work.

Box 1. Data, definitions and limits

Cohort. The 7 April 2026 PRIDE snapshot contains 36,249 public datasets with release dates between 2013 and 2025 (Supplementary Table S1). Release year means the year when a dataset became public in PRIDE. We state explicitly when an analysis uses submission year instead.

Units. Results are reported per dataset unless stated otherwise. Papers are publication records in OpenAlex[36], a catalogue of scholarly works, matched to the publication records of PRIDE datasets. Contributor groups are defined by hashed contacts of principal investigators. A first-time group has no earlier PRIDE dataset released since 2013. When a dataset names several instrument models, tools or species, its credit is divided equally among the named categories within each of these dimensions.

Effective number. Hill numbers[37] express diversity as the number of equally common categories that would have the observed diversity. They are calculated from category shares p_i among datasets. We use $1/\Sigma p_i^2$ at order 2, which gives more weight to common categories. At order 1 the expression is $\exp(-\Sigma p_i \ln p_i)$. Order 0 measures richness, the number of distinct categories. We rarefied richness by comparing equal numbers of datasets[38].

File evidence. We examined top-level file lists (3.03 million files) and the members of 45,985 results archives. Archive members were listed from zip central directories, which index the files in an archive, using HTTP requests for selected byte ranges. We matched column headers in deposited tables to characteristic outputs of analysis tools. We also extracted software versions from parameter files, logs and workflow files, and dates from raw and database file names. Each source provides incomplete evidence of file content, so detection gives a lower bound on its presence.

Standard result files. These are submitter-deposited mzIdentML or mzTab files, including files identified by expanded name matching. Files generated by the repository are excluded.

Uncertainty. We repeatedly sampled whole contributor groups with replacement to obtain bootstrap distributions. Their percentiles supplied 95% intervals (seed 20261001; 200 to 2,000 replicates depending on the analysis). Contributor-group effective numbers are descriptive, without bootstrap intervals. Mention windows use calendar years and do not resolve within-year chronology.

What the data cannot show. The records do not establish instrument ownership or purchase dates, laboratory size, the reasons for packaging decisions, who set a release date, or reuse that leaves no text-mined mention. Country is unresolved for 22% of datasets.

Methods

Data sources. The analysis used a frozen snapshot documented in `data/Data-manifest.json`. Sources were PRIDE archive tables (projects, files, instruments, species and publications), ProteomeXchange registry links, OpenAlex records of linked papers (works, authorships and institutions) and Europe PMC records (journal titles and accession mentions).

Reading of files. We listed the files inside zip archives, including zip64 archives, by reading their central directories with requests for selected byte ranges. We read column headers from the first bytes of deposited text and xlsx tables and matched them to characteristic column patterns in the outputs of analysis tools. We extracted software release strings from parameter files and logs. Archive members were classified with the same name-based rules as top-level files. The header patterns used for recognition underwent model-assisted review and revision. This review did not provide an independent estimate of classification accuracy. Recognition of a file type does not verify its contents or usability. A repeat retrieval of the frozen targets on 11–12 October 2026 recovered 46,147 archive listings and 59,227 table-header records. Reclassification changed the final recognition outcome for one dataset; all annual recognition percentages agreed with the frozen analysis at one-decimal precision. The repeat retrieval is supplied as a sensitivity check, while the manuscript retains the original estimates.

Runs. Raw runs are mass-spectrometry acquisitions stored in vendor raw formats and identified from file names and extensions. Bundled formats were counted as single runs, so run counts are lower bounds. Exact counts are reported separately.

Instrument models. We harmonised instrument labels to model names. We excluded labels naming only a series or vendor, or no instrument, from model-level analyses.

Statistical uncertainty. We obtained 95% percentile intervals by bootstrapping contributor groups. For categories other than contributor groups, we calculated period ratios of effective numbers by resampling contributor groups independently within each period. Effective numbers of contributor groups and their period ratios are descriptive point estimates without bootstrap intervals. Resampling groups while retaining their category identities biases this concentration statistic.

Comparisons. Shift-share decompositions separate changes within categories from changes in the proportions of datasets in those categories. Rates of qualifying accession mentions were directly stan-

dardised: we weighted rates within release-year × human-status strata to match their shares in the full cohort. The expected first appearance of instrument models by continent was obtained by drawing model labels independently of continent within each release year.

Accession-mention outcome and model. For datasets released in 2013–2022, we modelled whether a qualifying non-primary paper mentioned the accession in the release calendar year or the next three calendar years. Qualifying papers named deposits from at least two contributor groups and no more than 100 datasets. We could not determine whether a same-year mention preceded or followed the dataset's release. A binomial logistic model estimated the probability of a qualifying mention. It used a ridge penalty (10^{-4}), which shrinks coefficients towards zero, and included standard result files, detected data-independent acquisition (DIA), sample and data relationship format (SDRF) files, human samples and a Thermo Fisher-only assignment from instrument labels. The model also adjusted for categorical release year, grouped software family and research field, log(1 + citations), a missing-citation indicator and $\log_{10}$(dataset bytes). The model included 20,684 datasets, of which 483 had a qualifying mention, from 8,363 contributor groups. Percentile intervals used 200 resamples of whole contributor groups. Supplementary Methods specify the transformations, treatment of missing values and reference levels. Sensitivity analyses reported unadjusted rate ratios for a restricted cohort with no DIA detected. A separate sensitivity analysis extended the mention window to the release calendar year plus five subsequent calendar years.

Sensitivity analyses. Each principal result was re-examined using alternative definitions, units of analysis and subsets. The outcome of each check (supported, weakened or refuted) and the resulting wording are recorded in Supplementary Table S21.

Computational implementation. Data queries used DuckDB[39]. Python analyses used NumPy[40], pandas[41] and SciPy[42]; the accession-mention model used statsmodels[43]. Data figures used Matplotlib[44]. The supplementary reference guide identifies the analysis programs, instrument models and file formats named in the text and figures.

Figures. Main and supplementary data figures were generated from code using frozen analytical outputs. Their captions state the relevant selection criteria, denominators and quantities estimated. Raster exports of the main and supplementary data figures have a width of 7.20 inches at 600 dpi. The unnumbered BioRender overview uses its native 2,752-pixel-wide preview, displayed at 7.2 inches (approximately 382 dpi).

AI assistance. OpenAI Codex assisted with code development and review, figure preparation, literature checking and manuscript drafting and editing. The author's final verification and approval of the analysis and text remain pending before submission.

Supplementary information

Supplementary Figures. 1, Concentration of datasets and bytes across contributor groups. 2, History of every common instrument model. 3, First appearance of new instrument models by continent. 4, Software version series in deposited files. 5, Age of software releases and sequence databases. 6, How the archive's bytes are distributed among file types. 7, File formats. 8, Isobaric labelling and post-translational modifications. 9, Organisms. 10, Countries. 11, International co-authorship. 12, Research fields of linked papers. 13, Timing of public release. 14, Submission timestamps.

Supplementary Tables S1 to S21. Supplementary-Tables.xlsx contains the tables; its contents sheet gives each table's definition and source.

The Supplementary Methods explain how mentions were identified, how the model was constructed and which inferences it supports.

Data and code availability

Code, derived inputs and figure evidence are at github.com/filiprumenovski/pride-data-economy. Offline checks reproduce the main run-count bootstrap, adjusted mention analysis, deposit-clock intervals and selected summaries. Fresh archive listings and table headers were retrieved on 11–12 October 2026; their checksums and comparison with the frozen analysis are supplied with the release. Contributor labels are pseudonymised, and the mapping back to contributor identities is excluded. Code is licensed under MIT; author-created data, text and figures use CC BY 4.0, excluding third-party assets such as the BioRender graphical abstract. The Zenodo archive is pending. Upstream geography inference and some historical sensitivity analyses are not reconstructed by the public scripts. The repository documents the checks performed and these remaining limits.

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Funding. This research received no specific funding.

Competing interests. The author declares no competing interests.

Author contributions. Pending confirmation of the contribution statement and final approval.

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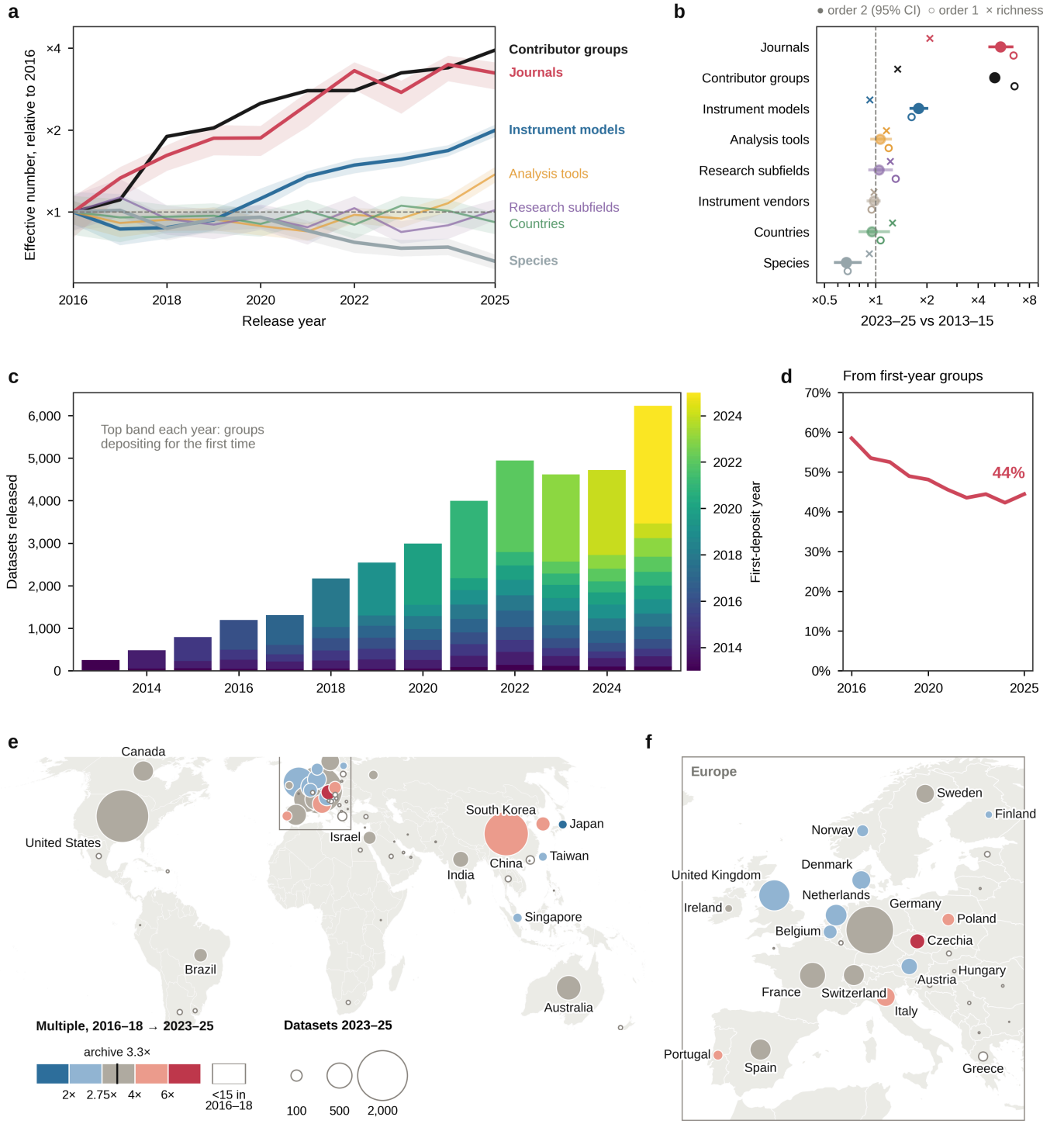
Main figures



g **The 20 largest contributors, 2023–25: datasets in each period (log scale) and growth multiple with 95% group-bootstrap interval**

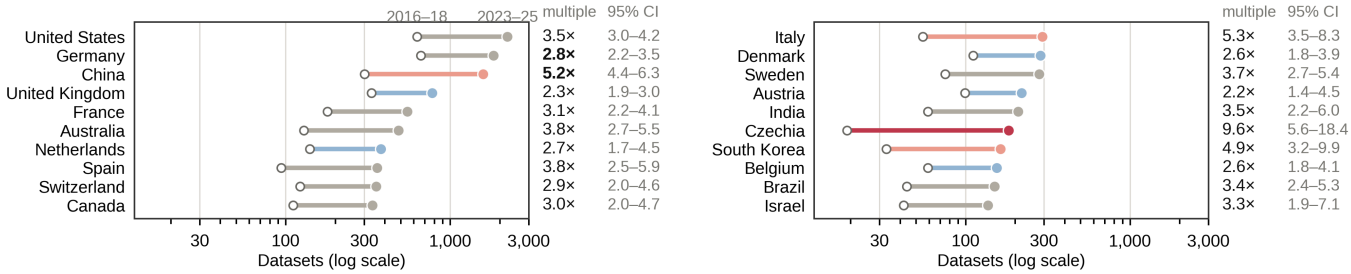


Figure 1 | An expanding community contributing to PRIDE. a–b, Diversity increased among contributor groups (×5.0), journals (×5.4) and instrument models (×1.8), and decreased among species (×0.67). Effective number (Hill number) combines the number of categories with their relative shares among PRIDE datasets. a, order 2 (inverse Simpson) by release year relative to 2016. b, 2023–25 vs 2013–15 at orders 2, 1 and 0 (category counts adjusted to equal dataset counts); 95% intervals from resampling contributor groups, except for contributor-group diversity, which is descriptive. A dataset naming several categories contributes equally to each. **c–d,** Four in ten public datasets come from groups depositing for the first time. PRIDE datasets per release year, grouped by the year the contributor group first deposited; groups are defined from hashed principal-investigator contacts. **e–g,** Europe supplied half of PRIDE's 3.3-fold growth; dataset counts grew 5.2× for China and 2.8× for Germany. Circle area: datasets released 2023–2025, by the contributing group's country; colour: multiple of the 2016–2018 count (archive-wide 3.3×). 3,632 and 12,085 datasets with an assigned country (78% of each period; the rest have no country assignment). Datasets added: Europe 4,218 (50%), North America 1,852 (22%), Asia 1,848 (22%; China 1,283). Counts describe deposits, not instrument ownership or national research output.

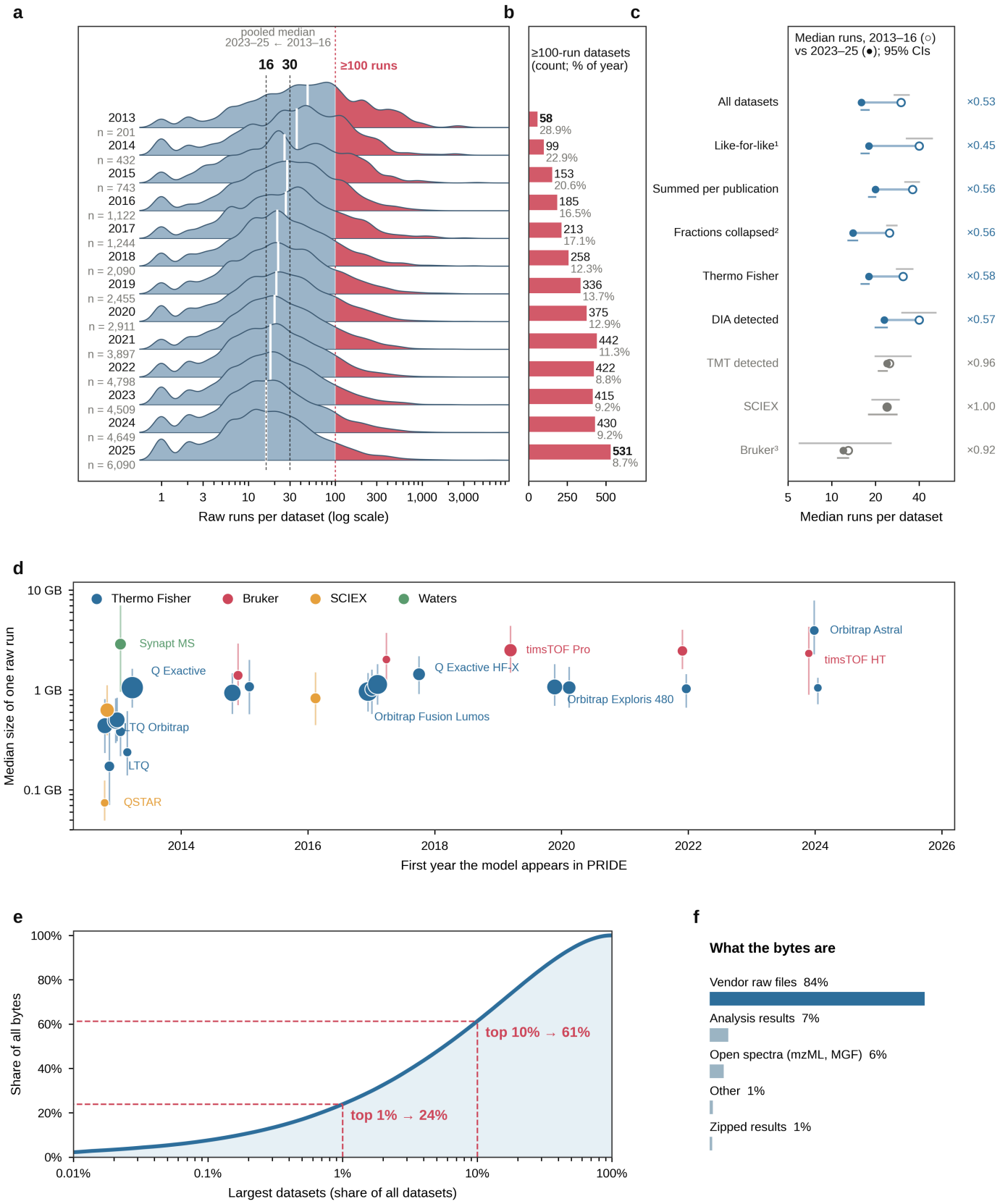


Figure 2 | A growing archive spanning different experimental scales. a–c, Typical datasets hold half as many raw runs as in 2013–16, yet the number with ≥100 runs rose from 58 in 2013 to 531 in 2025. PRIDE datasets released 2013–2025 with ≥1 identified raw run or raw bundle (n = 35,141 of 36,249). Runs are instrument acquisitions stored in vendor raw formats; bundled files make these counts lower bounds. Runs are not samples: TMTpro-18plex tandem mass tags can combine 18 samples in one run. Ridges: distribution per release year (equal area, log axis); white tick = median. Pooled medians 30 (2013–16, n = 2,498) and 16 (2023–25, n = 15,248); 95% intervals from resampling groups defined by hashed principal-investigator contacts. ¹ Thermo Fisher, no detected data-independent acquisition (DIA) or TMT, human samples, linked publication. ² Runs with fraction numbers combined into one count as an approximation to sample count. ³ Bruker 2013–16 n = 56. Grey: the confidence interval for the ratio includes 1. **d,** Median raw-run size is 9× larger for the Orbitrap Astral than for the LTQ Orbitrap. Median size of one raw run by instrument model (bars: interquartile range), placed at the model's first year in PRIDE. **e–f,** 1% of datasets hold 24% of the archive's 2.71 petabytes. Cumulative share of PRIDE bytes held by the largest datasets (all deposited files, 2013–2025).

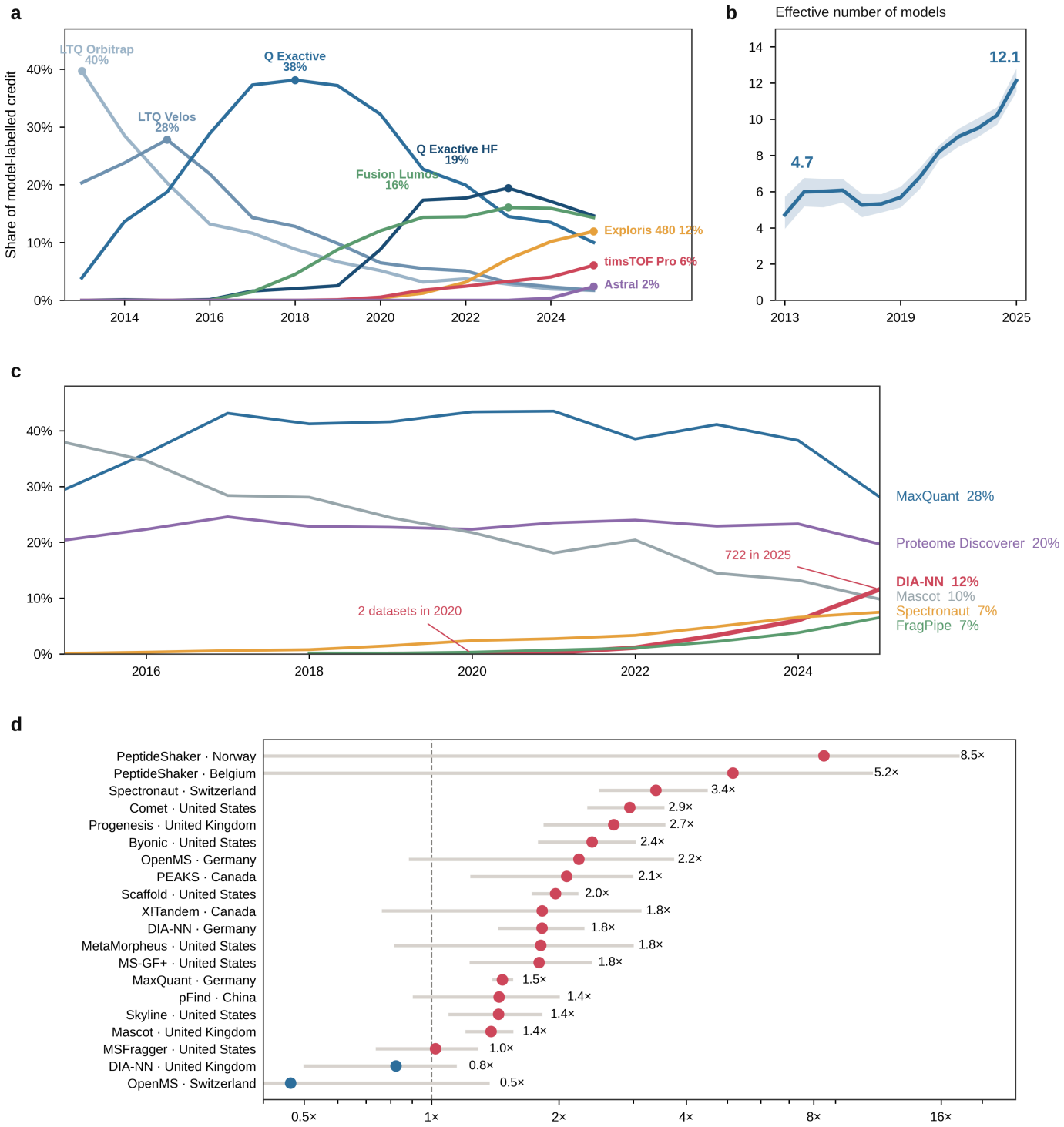


Figure 3 | A public record of instruments and analysis workflows. a–b, Deposits named new instrument models alongside older ones. a, Share assigned to each selected model among datasets with an instrument-model label; a dataset naming several models contributes equally to each. Every model named in ≥50 datasets still appeared in 2025 releases; deposits include delayed releases and reanalyses. b, Effective number of models (inverse Simpson index of the weighted model shares); shading: 95% intervals from 300 bootstrap resamples of contributor groups. **c,** Within five years, DIA-NN became the third most frequently named analysis program in public datasets. Share of PRIDE datasets naming each analysis tool (protocol text or software field), by release year. **d,** Software is named more often in deposits from its developers' country: 18 of 20 tool–country pairs. Frequency of each program among groups in its developer's country relative to all countries combined (1 = equal frequency); 95% intervals from resampling contributor groups. Median 1.8×; permutation p = 0.0002.

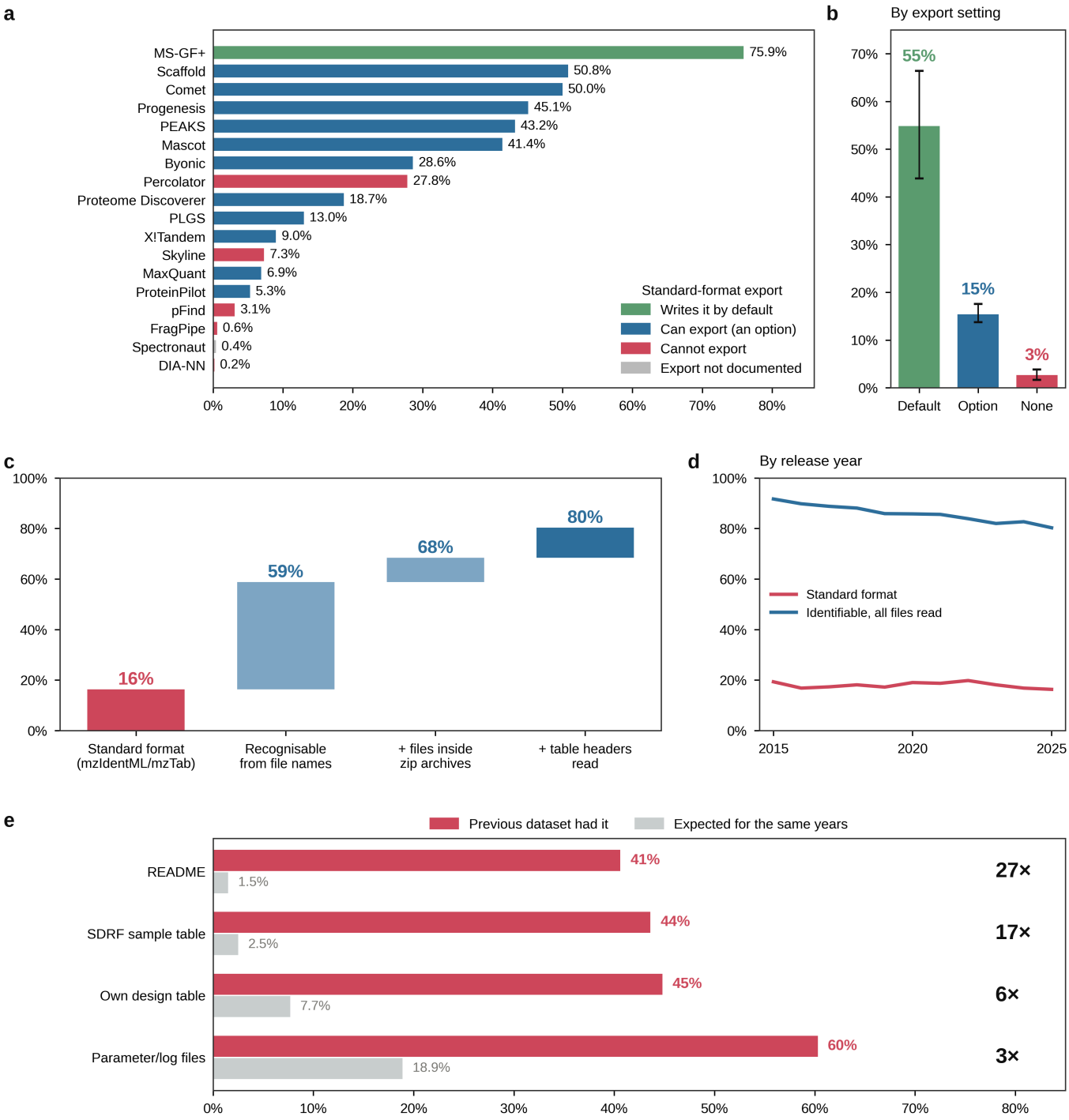
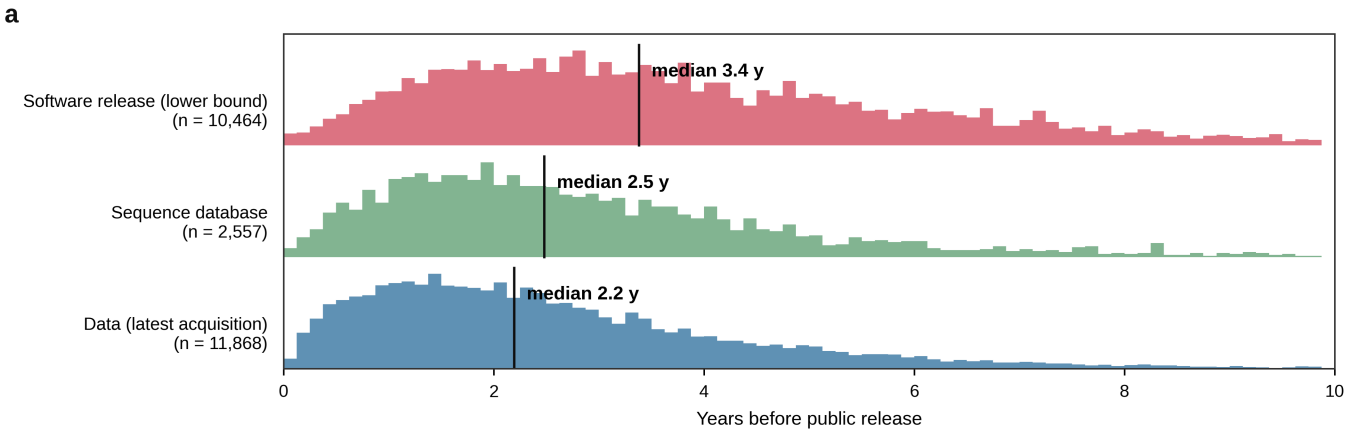
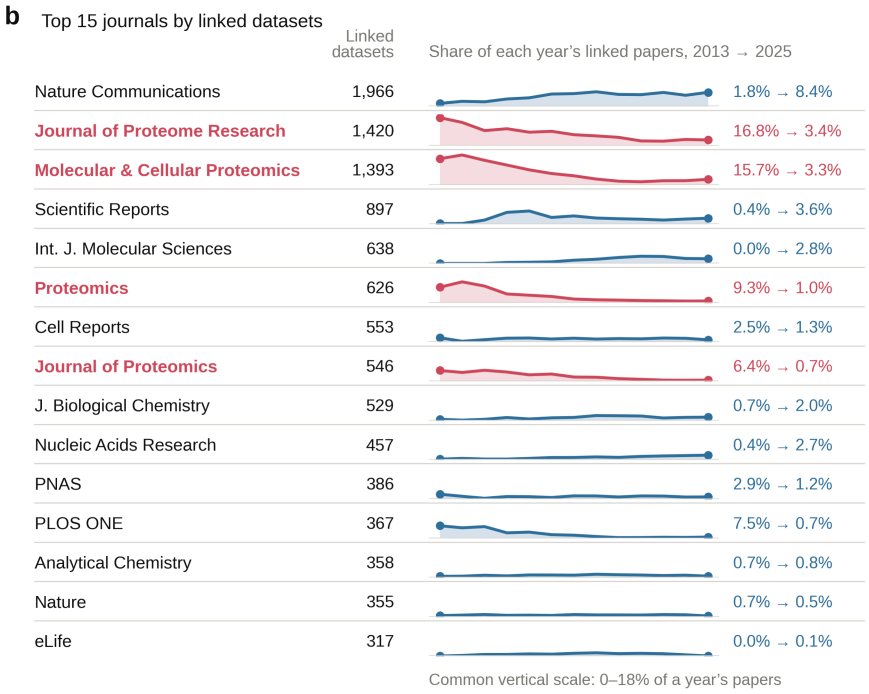
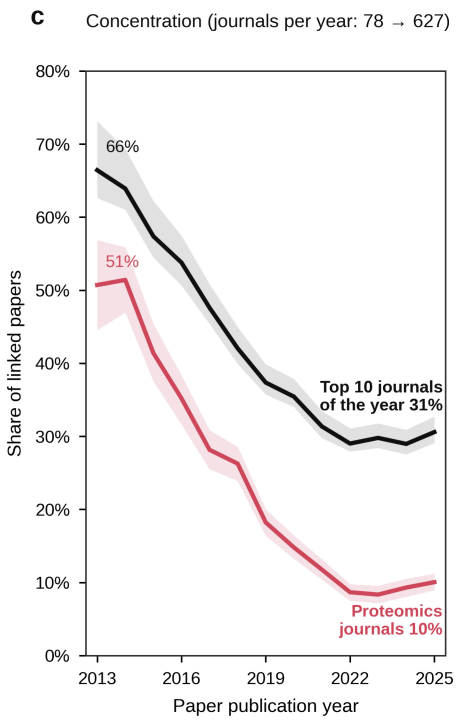
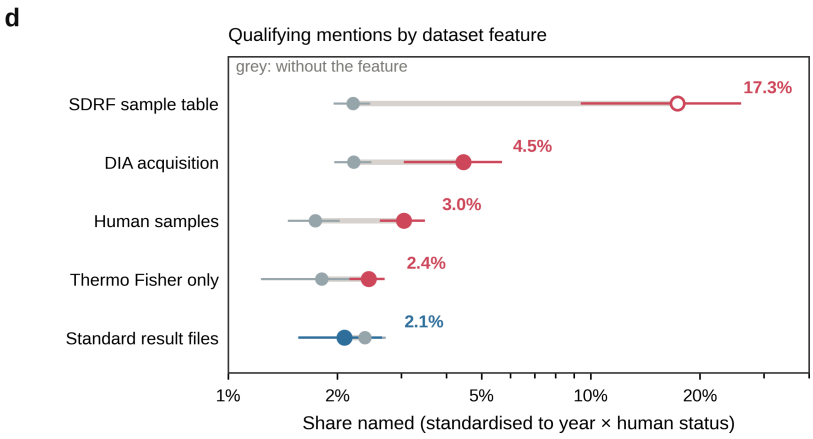


Figure 4 | Results shared in standard and software-specific formats. a–b, The share of datasets with standard result files ranges from 0.2% to 76% across analysis programs. PRIDE datasets naming a single analysis program and containing submitter-deposited mzIdentML or mzTab files; programs with ≥60 datasets. Repository-generated files are excluded. **c–d,** Reading deposited outputs reveals identifiable results in 80% of 2025 datasets; 16% contain standard result files. PRIDE datasets released in 2025 (n = 6,231) with identification or quantification results recognised from file names, archive contents and table headers. **e,** README documentation and sample-to-file mappings in SDRF files recur within contributor groups more often than expected from their overall frequencies. Share of a group's next PRIDE dataset carrying each file when the previous one did, vs the rate expected for datasets of the same years.









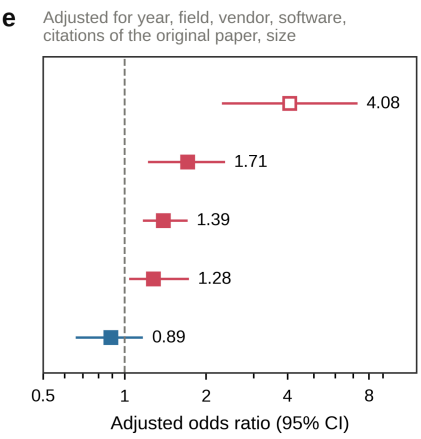


Figure 5 | Connecting public deposits to the research literature. a, At public release, the median time since acquisition was 2.2 years and the software release was at least 3.4 years old. Age at public release of the latest raw-file acquisition date (datasets with dated raw-file names, 34% of those with raw files), the sequence database date and the software release (first seen in PRIDE; lower bound). The plot excludes one negative database interval and two negative software intervals; ages above 10 years are pooled in the final bin. **b–c,** Proteomics journals carried 51% of dataset-linked papers in 2013 and 10% in 2025. Journals of linked papers, obtained from Europe PMC using PubMed identifiers: 22,475 papers published 2013–2025 (525 without a journal, mostly preprints, excluded). Journals ranked by linked datasets; small line plots show each journal's share of papers per year, counting each paper once. Proteomics journals = title contains "proteom" (18 titles). Bands: 95% bootstrap over contributor groups. **d–e,** Qualifying papers more often name human datasets and datasets with data-independent acquisition (DIA); standard result files show no detectable association. PRIDE datasets released 2013–2022 (n = 20,684) mentioned by identifier in a qualifying paper other than their primary reference, during the release calendar year or the next three calendar years (2.3% overall). Qualifying papers name datasets from ≥2 contributor groups and no more than 100 datasets; publication and release order within a year is unresolved. Thermo Fisher only means a single-vendor assignment from instrument labels. Rates are standardised to release year and human status; odds ratios use the adjusted model in Methods. Europe PMC text searches miss some dataset mentions, and a mention does not establish that the data were analysed again. Open circle: SDRF files were often added after release by reuse projects (91 of 205 datasets predate the format), so the association's direction cannot be determined.

Supplementary methods

Mention ascertainment. We excluded pairs of PRIDE accessions and PubMed identifiers (PMIDs) that were marked as primary references. The main outcome required a mentioning paper to name no more than 100 datasets spanning at least two contributor groups. We computed these counts for each paper before restricting mentions to publication year minus release year of 0–3 inclusive. A paper was not excluded solely because it was the primary paper for another deposit from the same contributor group. Same-year mentions can precede the exact release date. The sensitivity analysis with five subsequent calendar years used publication year minus release year of 0–5 among 2013–2020 releases (11,741 datasets). The restricted cohort comprised 4,399 human datasets assigned solely to Thermo Fisher from instrument labels, with no DIA detected, naming MaxQuant or Proteome Discoverer. Neither restriction establishes raw-file presence or confirms data-dependent acquisition (DDA), in which precursor selection depends on signals in the acquired survey spectra.

Model construction. We fitted a binomial generalised linear model (GLM) with an intercept in statsmodels[43], using its logit link and `fit_regularized(alpha=1e-4, L1_wt=0)`. The logit link relates the predictors to the log odds of a qualifying mention. Missing primary-publication citation counts (5,556 datasets) were replaced with zero before the natural-log transformation log(1 + citations). A separate indicator recorded which citation counts were missing. This indicator does not prove absence of a publication. For each dataset, the citation count was the maximum count reported by iCite[45], the citation-data source, across its linked publication records. Listed byte counts below 10^6 bytes were set to 10^6 before the $\log_{10}$ transformation. Missing values would also be set to this floor. No size values were missing in the model cohort, and 25 fell below the floor. Missing research fields were coded as "None" (4,460 datasets). The eight most frequent resulting categories were retained; all others were pooled as "Other". The seven most frequent software-family categories were retained and the remainder pooled. Datasets naming multiple major tools formed a distinct software group. The reference levels were 2013 for release year, Mascot for software family, and Agricultural and Biological Sciences for research field. Human-sample and SDRF flags were set to false when those features were not detected. Missing contributor-group identities would receive accession-specific identifiers, although this fallback was not needed in the model cohort. All 20,684 dataset rows entered the model's design matrix of predictor values, which had 33 columns including the intercept.

Intervals and estimands. The logistic-model point estimates used every dataset in the cohort. Two hundred cluster-bootstrap fits resampled whole contributor groups with replacement, keeping the same predictor columns and coding. We exponentiated the coefficients to obtain odds ratios and used their 2.5th and 97.5th percentiles as interval endpoints. The seed was 20261001; the source script shares its random-number stream with preceding analyses, so the execution order affects the exact bootstrap sequence. An offline rerun exactly reproduced the frozen results: all 201 fits had finite coefficients and no fit warnings. Maximum absolute penalised scores, which measure the remaining gradient of the penalised objective, were below 1.0×10^{-5}. With 200 draws, random variation in the bootstrap limits the precision of the interval tails. The Thermo Fisher-only lower endpoint crossed one when the draws were split into two halves for a diagnostic check. Figure 5d shows rates directly standardised over release-year × human-status strata to the full cohort distribution, with 1,000 contributor-group bootstrap draws. Strata with no datasets in a subset or bootstrap replicate were omitted. The weights of the remaining strata were rescaled to sum to one. Figure 5e shows adjusted odds ratios. Sensitivity analyses report unadjusted ratios of the mention rate in datasets with each feature to the rate in datasets without it, using 1,000 contributor-group bootstrap draws.

Descriptive diversity. Contributor-group effective numbers are reported without intervals. In the original bootstrap, repeated draws of a group retained the same category identity, changing category concentration. Effective numbers for other categories retain the contributor-group bootstrap intervals from the frozen analysis. Rarefaction uses the frozen pooled values underlying the figures. Subsequent reruns use a fixed ordering of datasets and categories. The instrument richness comparison rounds to 61 versus 66.

Supplementary reference guide

These references identify the software, instruments and formats named in the text, figures and software registry. They provide technical background and credit to their developers. Dataset counts, shares and comparisons come from this study. A reference to a software family does not identify the version in every deposit or establish its export capabilities. The instrument table retains the recorded model labels and states where a source covers only a family.

Analysis software

Software or grouped label	Reference; qualification where needed
MaxQuant	21
DIA-NN	20
Proteome Discoverer	22 Platform overview.
Spectronaut	23
Comet	24
Mascot	28
MS-GF+	27
MetaMorpheus	29
FragPipe / MSFragger	46,47 Platform and search-engine references; the platform paper is later than the study period.
Skyline	48
PEAKS	49
Scaffold	50
pFind / pLink	51,52 Separate references for the two named programs.
OpenMS / OpenSWATH	53,54 Separate references for the two named programs.
Byonic	55
X!Tandem	56
PeptideShaker / SearchGUI	57,58 Separate references for the two named programs.
SEQUEST	59
ProteinPilot	60 Paragon search algorithm used in ProteinPilot.
Percolator	61
Progenesis	62 Proteomics software family; official documentation.
PLGS (ProteinLynx Global SERVER)	63 Manufacturer documentation.
Sage	64
EncyclopeDIA	65
TPP (Trans-Proteomic Pipeline)	66

Instrument models

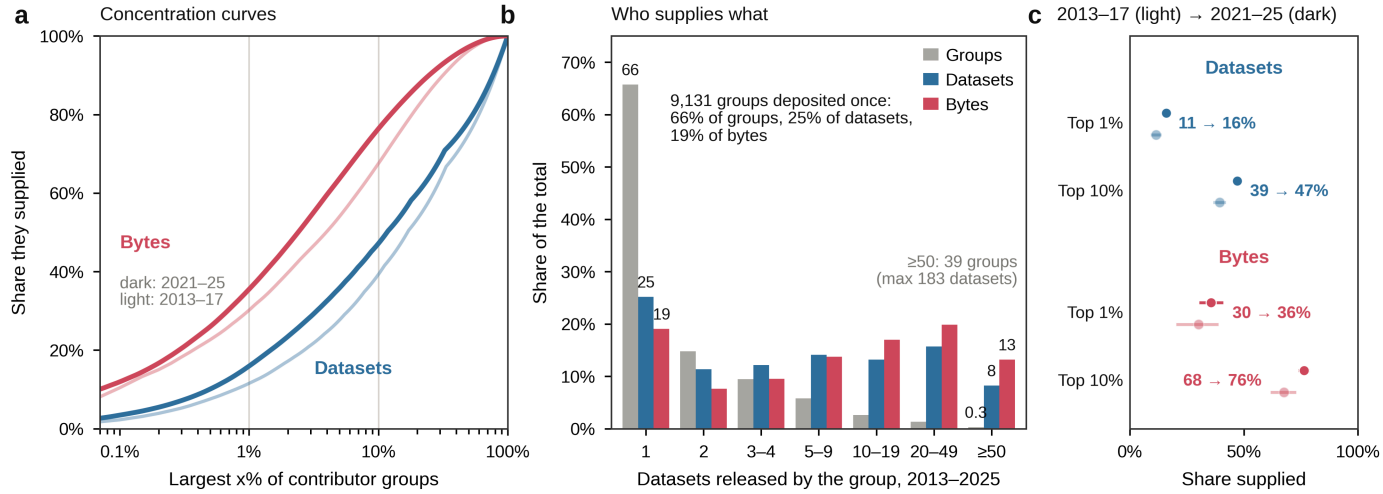
Recorded model label	Reference; qualification where needed
LTQ Orbitrap	14
LTQ	67
LTQ Orbitrap XL	68
4800 Proteomics Analyzer	69
QSTAR	70
LTQ Orbitrap Velos	71
TripleTOF 5600	72
LTQ Orbitrap Elite	73
LTQ FT	74
Q Exactive	16
Synapt MS	75 Synapt platform; the recorded label does not specify a variant.
6520A Quadrupole Time-of-Flight LC/MS	76 6520 family; cited catalog does not establish the A suffix.
ultraflex	77 LIFT-TOF/TOF family; generation is unspecified.
Q-Tof Ultima	78 Ultima Global manual; the recorded label is less specific.
Orbitrap Fusion ETD	79
maXis	80
Orbitrap Fusion	79
Q TRAP	81
TripleTOF 6600	82
impact II	83
Q Exactive Plus	84
Q Exactive HF	17

Recorded model label	Reference; qualification where needed
Orbitrap Fusion Lumos	85
Q Exactive HF-X	86
autoflex	87 Manufacturer documentation for the autoflex family.
impact HD	88
timsTOF Pro	18
SYNAPT G2-Si	89
Orbitrap Exploris 480	90
Orbitrap Eclipse	91
timsTOF Pro 2	92
Orbitrap Exploris 240	93
timsTOF	94 Original timsTOF; the deposited label may also be generic.
timsTOF SCP	92
Orbitrap Astral	15
timsTOF HT	92
Orbitrap Ascend	91

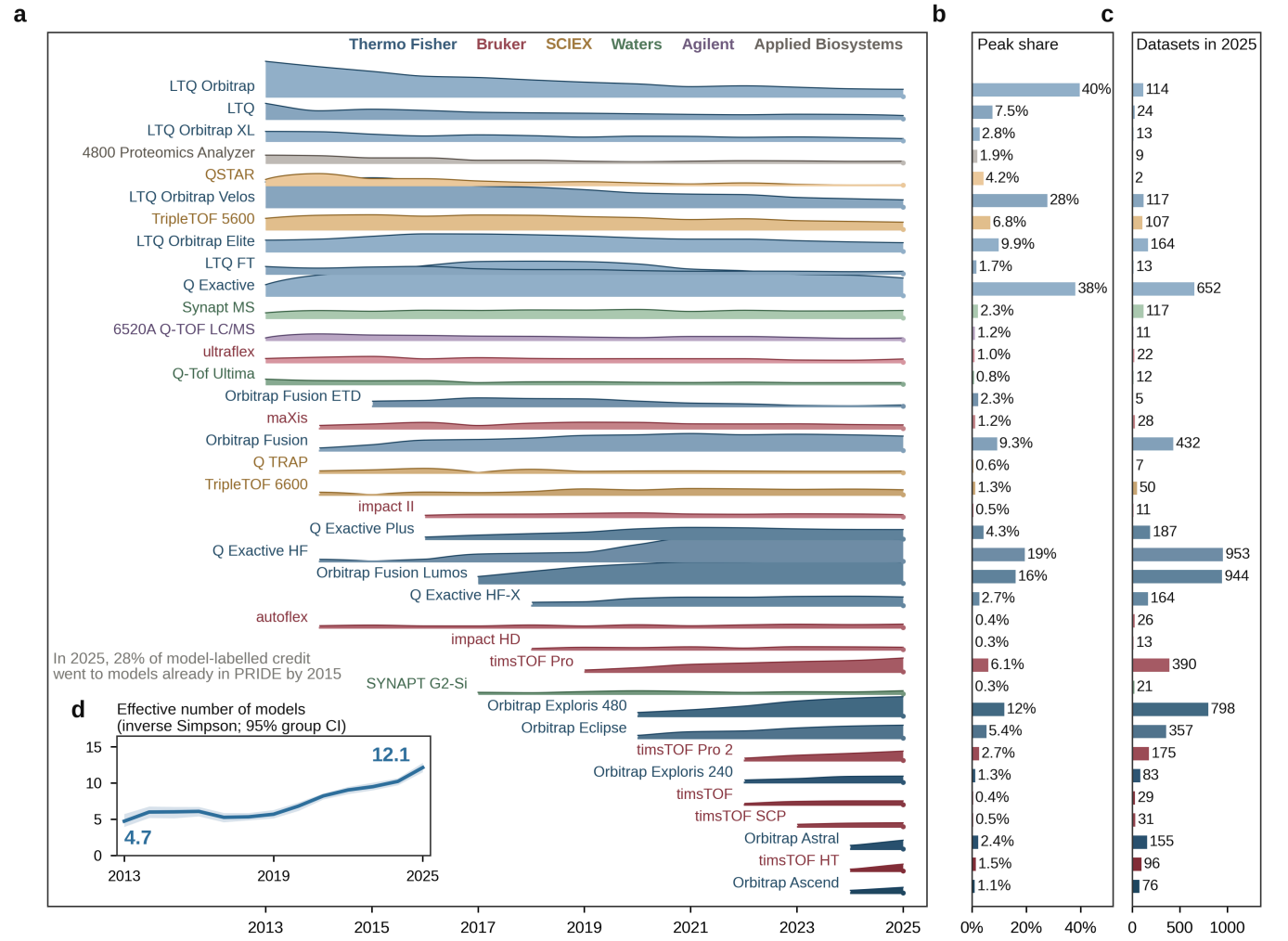
Labelling, acquisition and file formats

Method or format	Reference
TMT; TMTpro; TMTpro-18plex	13,11,95
iTRAQ	12
DIA/SWATH	30
mzIdentML; mzTab; mzML	25,26,96
SDRF sample-to-file metadata	32

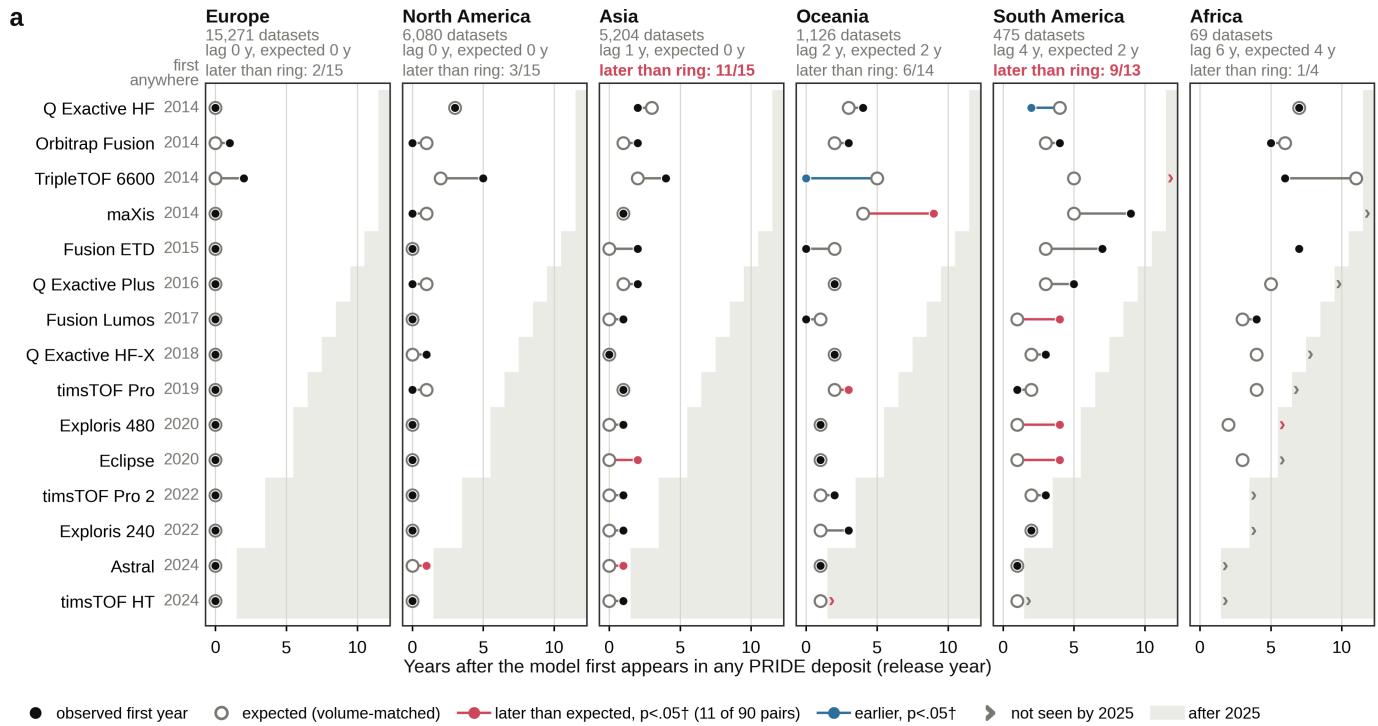
Supplementary figures



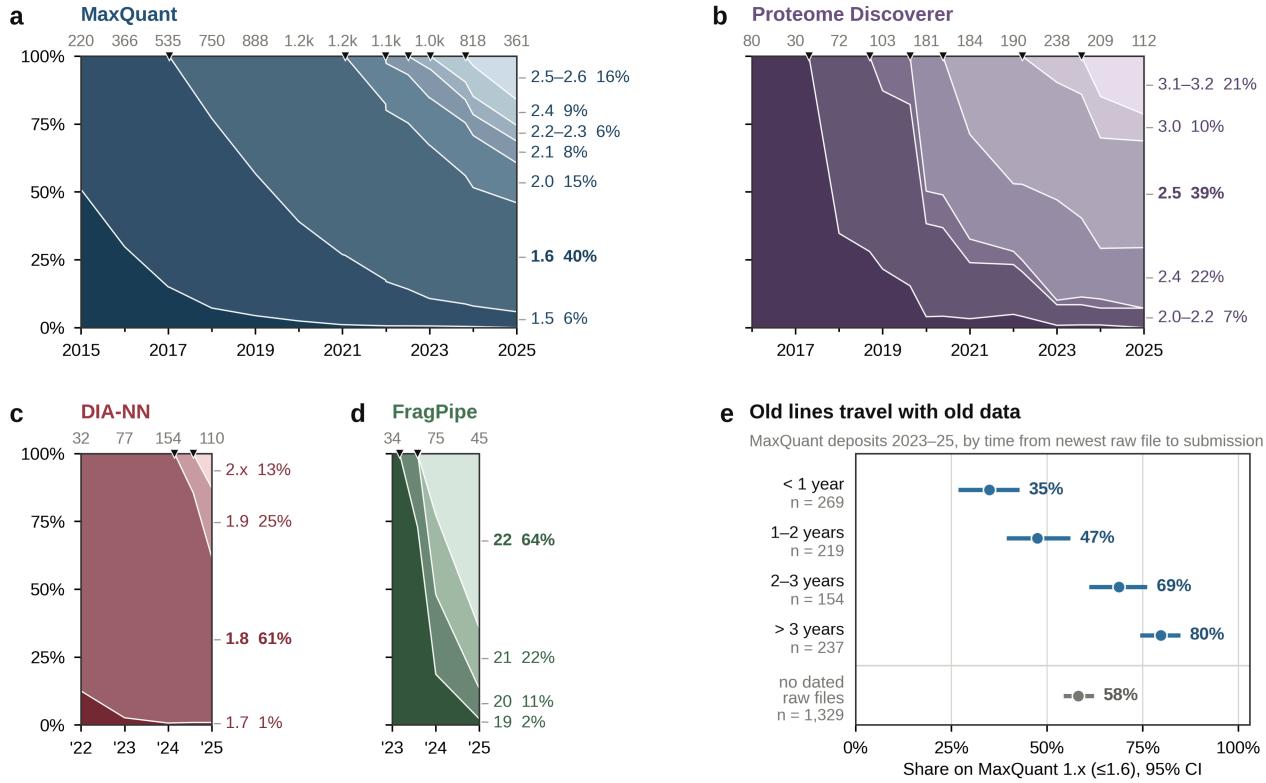
Supplementary Fig. 1 | Datasets and storage concentrated among contributor groups. a–c, The top 1% of contributor groups released 18% of PRIDE datasets and 38% of its bytes. 36,244 datasets released 2013–2025 from 13,892 contributor groups (hashed principal-investigator e-mail/ORCID; top 1% = 139 groups; 5 datasets without one principal investigator). Bytes = all files listed per dataset. Gini index of inequality in dataset counts: 0.42 in 2013–17 and 0.49 in 2021–25. Lines in c: 95% bootstrap over groups.



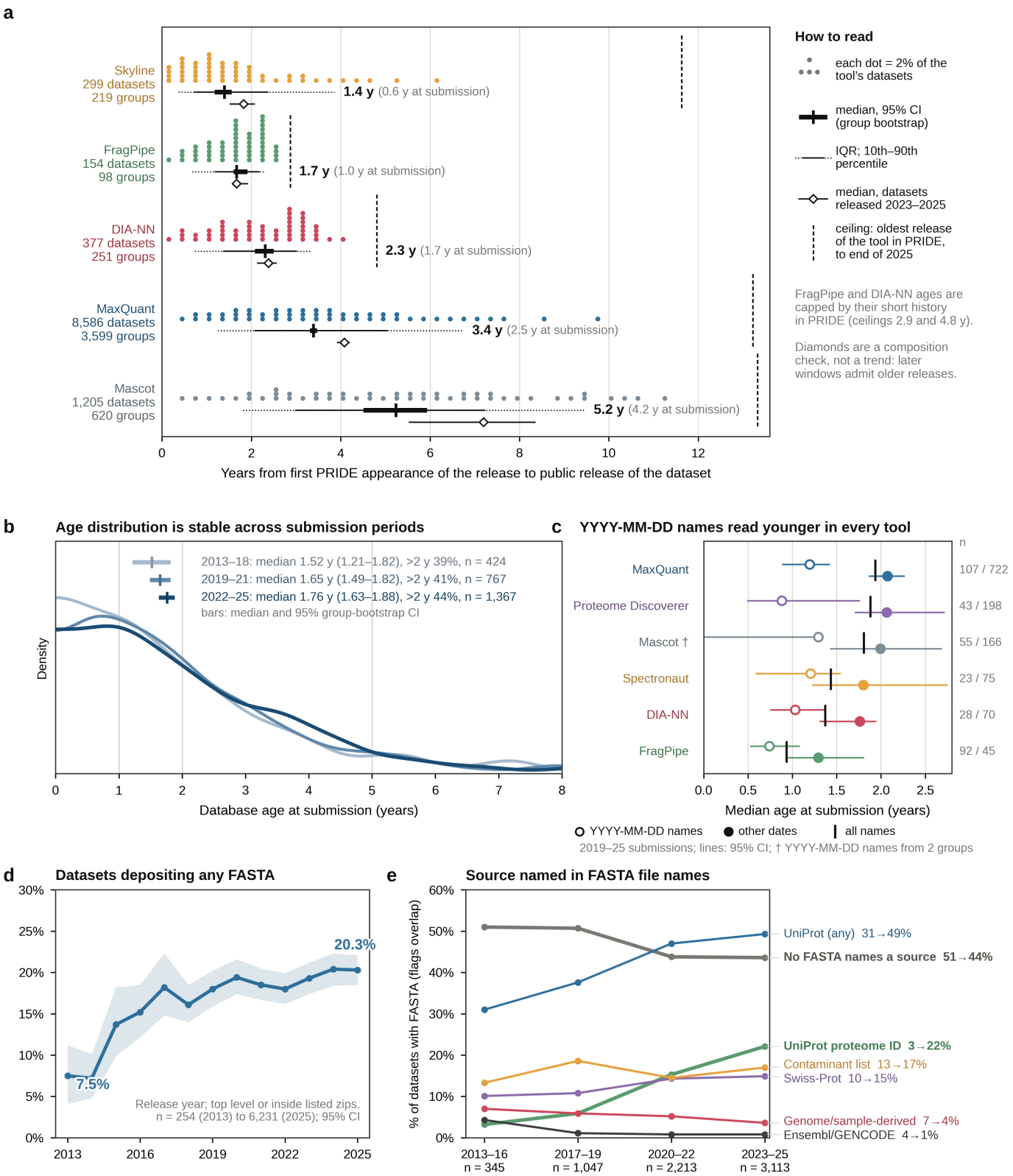
Supplementary Fig. 2 | Deposit history of common instrument models. a–d, All 37 common models still appear in 2025 deposits. Rows: models named in ≥50 public PRIDE datasets released 2013–2025 (35,770 datasets with model labels; each dataset contributes equally to its named models), ordered by first year with ≥3 datasets. Ridge height = √(share of that year's model-labelled datasets), common scale. Thin line = years with ≥1 dataset. For rows starting in 2013, earlier appearances fall outside the study period. An instrument label does not establish ownership; its first recorded year does not establish a purchase date.



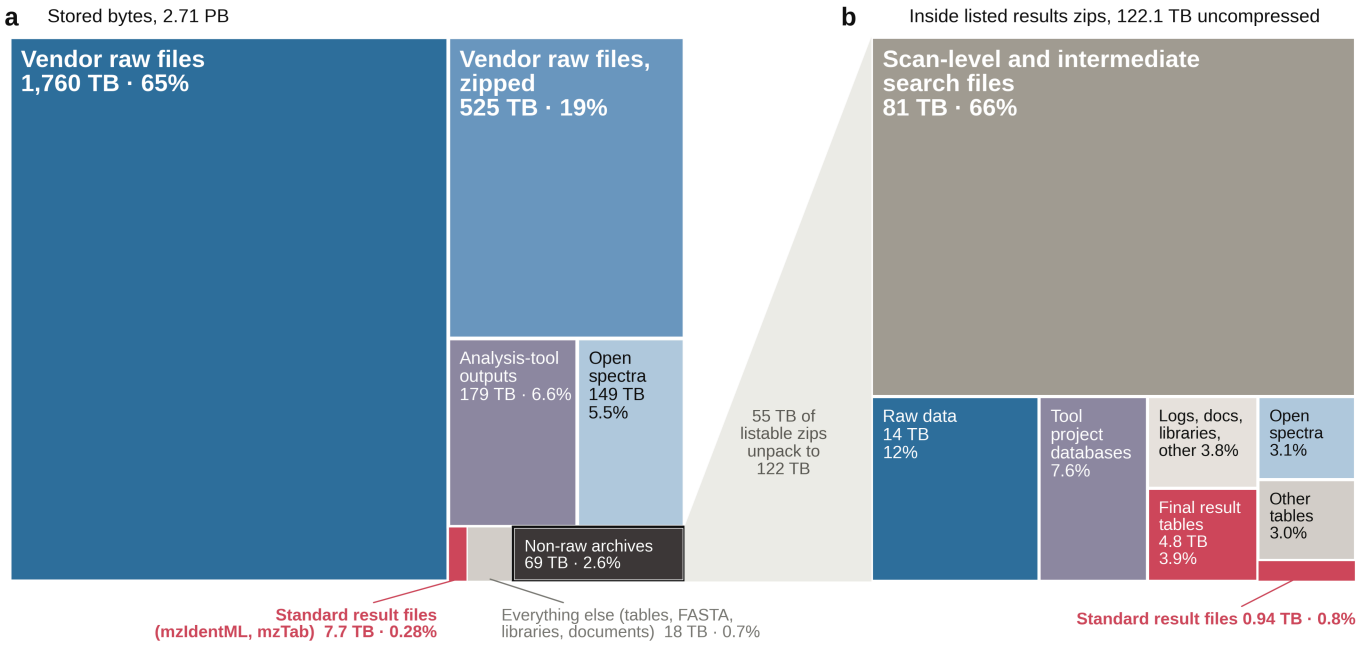
Supplementary Fig. 3 | First deposits naming new instrument models, by continent. a, Africa's 6-year delay is close to that expected from its 69 datasets. First release year of 15 models (≥100 datasets; first ≥5 in 2014 or later) in each continent, based on principal-investigator country (28,225 of 36,249 datasets). Delay is measured from each model's first deposit anywhere and summarised as the median across models observed. Ring: expected median first year if instrument labels were independent of continent within each year, given each continent's dataset count. A deposit is not a purchase record. † Hypergeometric comparison assumes independent datasets. It ignores clustering within contributor groups, making p-values too small (≈4.5 of 90 pairs expected below .05 by chance).



Supplementary Fig. 4 | Software version series in deposited files. a–e, Among 2025 MaxQuant deposits with a recoverable version, 40% used the 1.6 series, first seen eight years earlier. Share of PRIDE datasets recording each software version series, by submission year (datasets released 2013–2025). Versions are read from deposited parameter files, logs and workflows (MaxQuant, DIA-NN, FragPipe) or mzIdentML writer fields (Proteome Discoverer). Black triangles: the first record of a version series by a second contributor group. Grey numbers: datasets with a version. 2025 coverage: MaxQuant 361 of 576 datasets naming the tool, Proteome Discoverer 112 of 486, DIA-NN 110 of 331, FragPipe 45 of 142. a–d: submission (deposit) year on one time scale; 2025 = submissions public by 31 Dec 2025. e: 95% group-bootstrap intervals over hashed contributor groups.



Supplementary Fig. 5 | Ages of software releases and sequence databases. a, At public release, median software-release age is ≥5.2 years for Mascot and ≥1.4 for Skyline. PRIDE datasets released 2013–2025 whose deposited files record an exact software release (one value per dataset and tool). Age = public release date minus the first PRIDE submission recording the same release. Software exists on or before that date, so every age is a lower bound. Using the first record by a second contributor group instead gives median ages of Skyline 1.2, FragPipe 1.5, DIA-NN 2.1, MaxQuant 3.3 and Mascot 4.9 years. Excluded: 2 datasets whose PRIDE release date precedes the first-seen submission; 4 Skyline files whose "version" was "64-bit". **b–e,** Deposited sequence databases are typically 1.7 years old, and 42% are older than two years. PRIDE datasets released 2013–2025. b, c: age of the main sequence database (FASTA) at submission, from a high-confidence date in its file name (2,558 datasets, 1,305 groups; median 1.67 years, 95% confidence interval 1.58–1.80; >1 year 69%, >5 years 8%). A name date may be a download, release or tool build date, so ages are approximate.

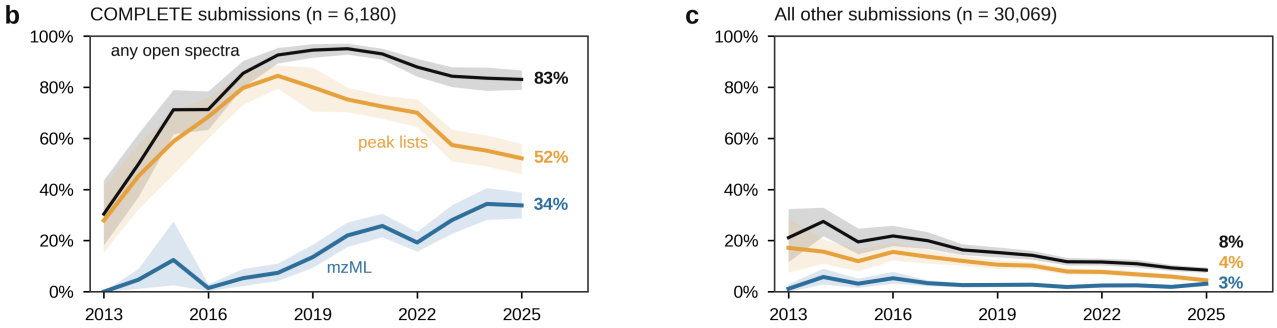


Supplementary Fig. 6 | Storage occupied by different file types. a–b, Raw data occupy 84% of PRIDE's 2.71 petabytes; final result tables occupy 4% of uncompressed results-archive storage. (a) Stored bytes of all 3.03 million top-level files in the 36,249 study datasets, by file category. (b) Uncompressed bytes inside 45,985 results zip archives with readable file lists (55.3 of 68.9 TB of non-raw archives; rar, 7z and tar were not listed), classified by file role from file names. The two panels use different scales.

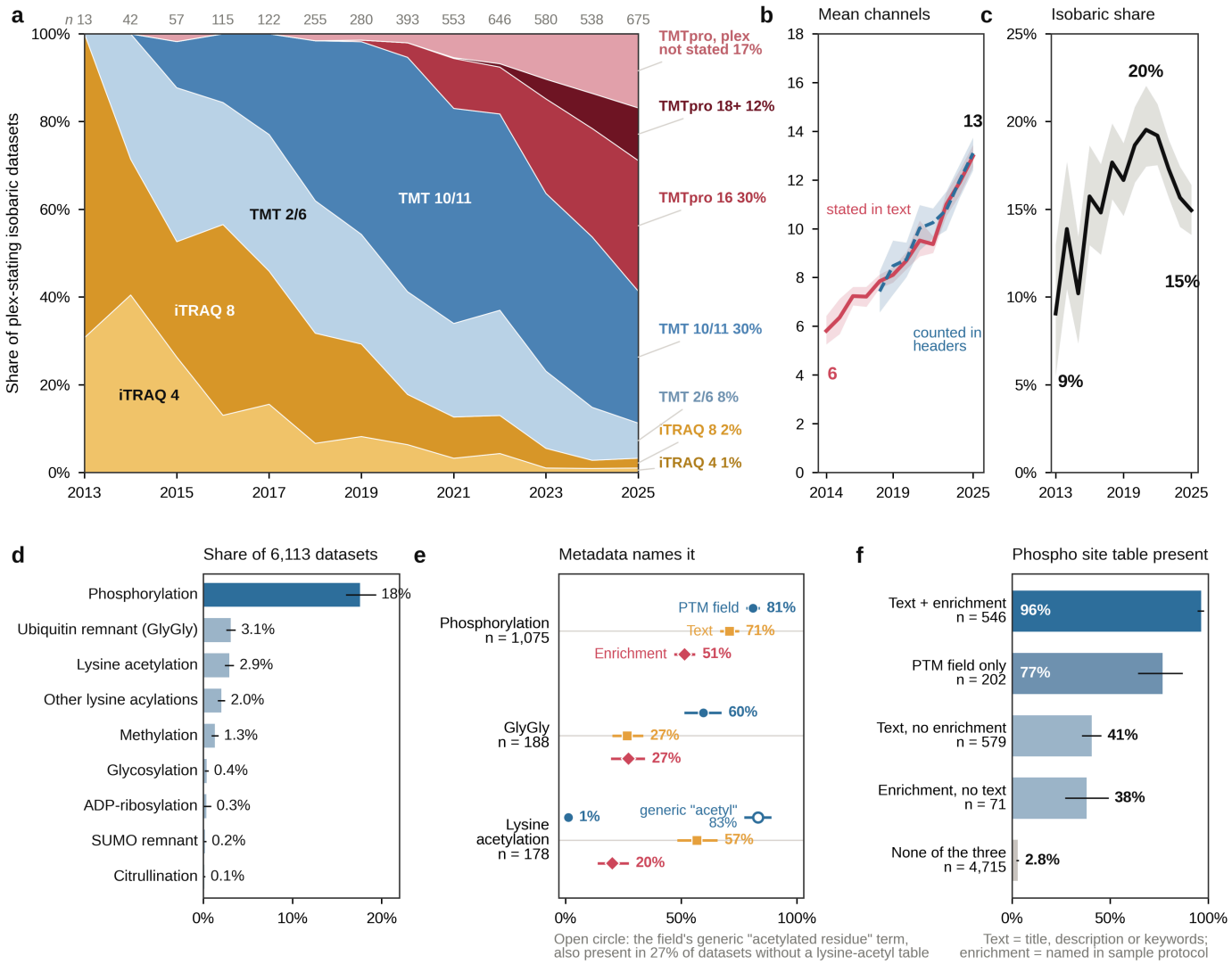
a

		2013	'14	'15	'16	'17	'18	'19	'20	'21	'22	'23	'24	2025	13–16 → 23–25 pp	
Vendor raw	**Any vendor raw**	**70**	**82**	**87**	**89**	**90**	**91**	**92**	**93**	**94**	**93**	**94**	**94**	**91**	85 → 92	+7
	Thermo .raw	67	73	78	82	84	84	85	86	88	85	85	84	78	78 → 82	+4
	Bruker .d	1.2	2.5	2.0	1.3	1.4	2.0	2.1	2.7	3.4	3.7	5.5	7.4	11	2 → 8	+7
	SCIEX .wiff	3.9	8.5	7.9	6.9	5.3	6.7	6.1	4.7	4.1	4.7	3.2	3.3	2.7	7 → 3	-4
Open spectra	**Any open spectra**	**24**	**35**	**36**	**33**	**31**	**30**	**27**	**27**	**25**	**25**	**23**	**20**	**20**	34 → 21	-13
	mzML	0.8	5.4	6.0	4.4	3.7	3.4	4.3	5.7	5.8	5.5	6.6	6.7	7.7	5 → 7	+2
	mzXML	2.8	6.2	5.3	3.3	5.0	3.1	4.2	2.8	3.0	2.1	2.9	2.2	2.2	4 → 2	-2
	Peak lists (MGF…)	20	26	27	28	25	25	21	20	19	19	15	13	12	27 → 13	-14
Results	**Standard results**	9.4	13	19	17	17	18	17	19	19	20	18	17	16	16 → 17	+1
	Tool-native outputs	31	31	28	32	31	35	36	41	43	42	44	45	44	31 → 44	+14
Context	FASTA	6.3	6.4	12	13	16	15	17	18	17	17	18	18	18	11 → 18	+7
	SDRF	0.4	2.3	2.8	1.5	2.0	0.6	0.9	0.8	0.5	1.0	2.0	3.0	3.5	2 → 3	+1
	README	0.8	1.7	0.3	0.6	1.0	0.5	0.7	1.1	0.7	0.6	0.9	0.7	0.9	1 → 1	+0

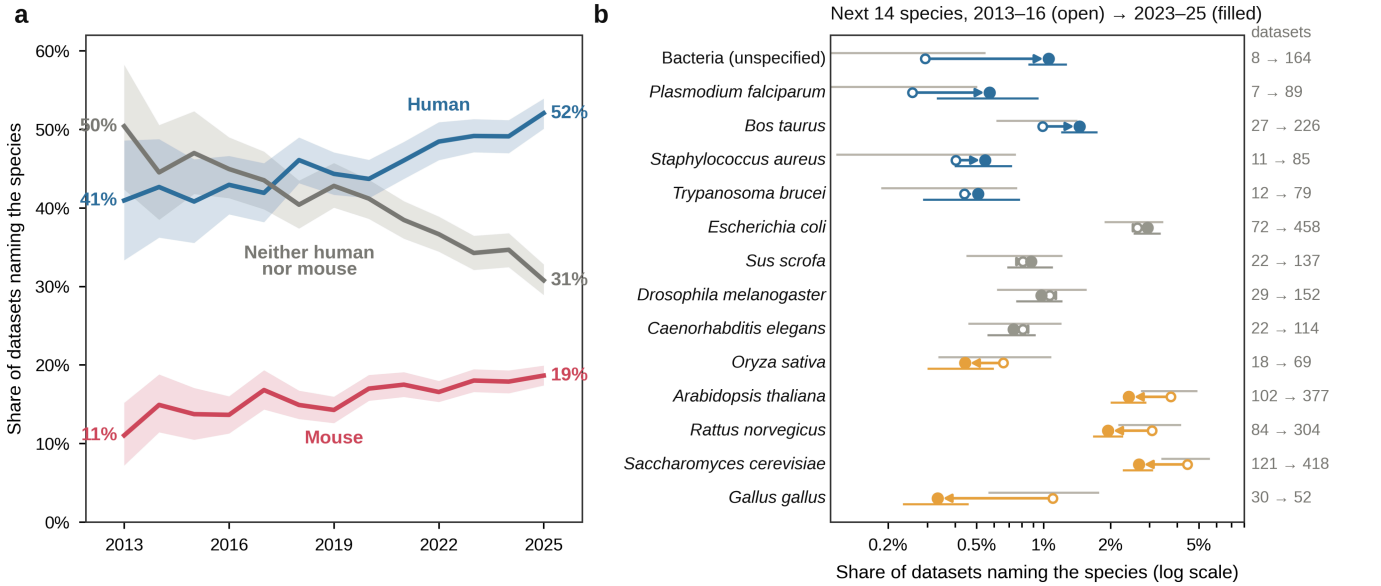
% of datasets with the file type, by release year; right: period shares and change (percentage points)



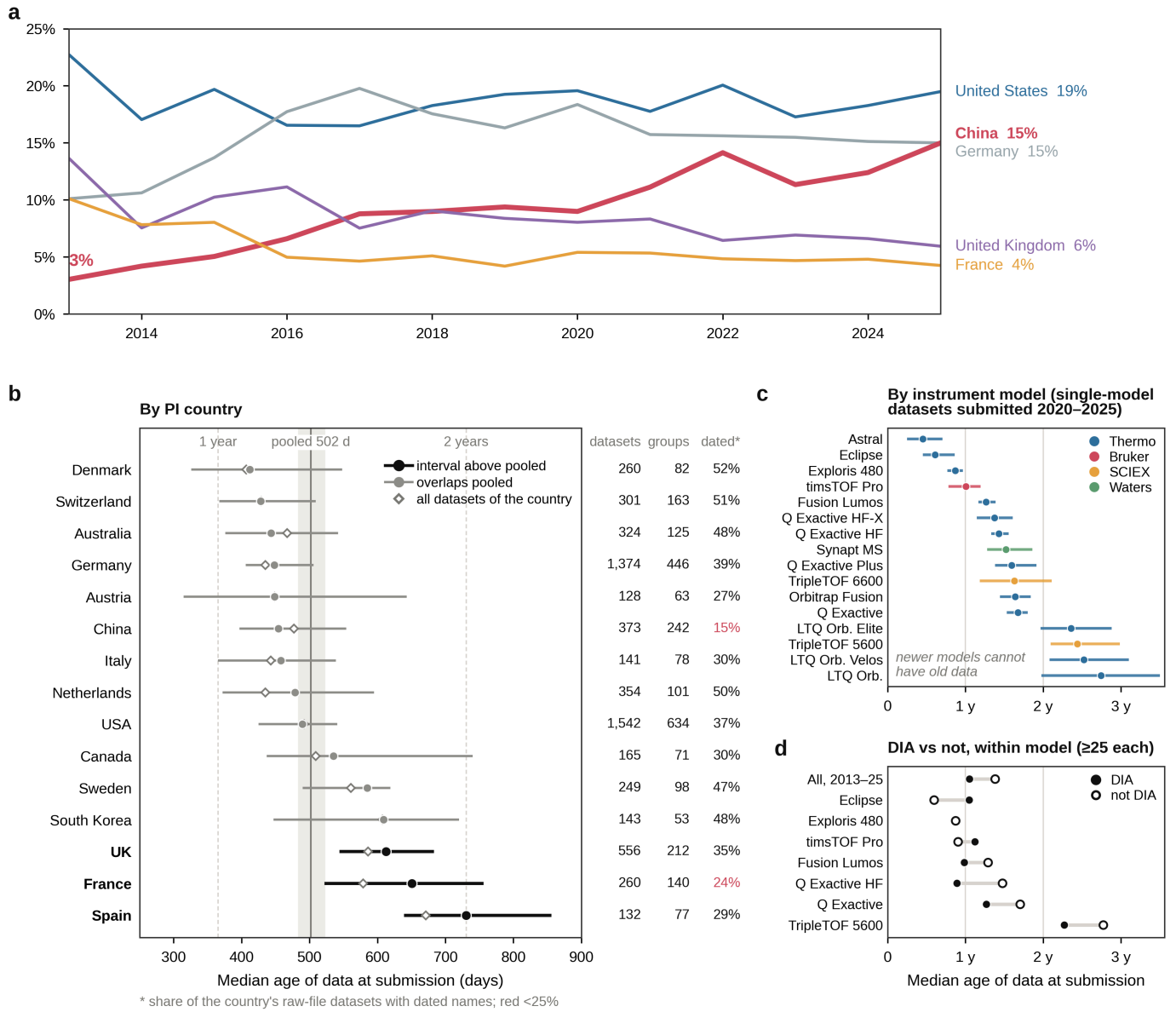
Supplementary Fig. 7 | File formats. a–c, Datasets with open spectrum formats fell from 34% to 21% as COMPLETE submissions halved; the share with mzML increased. PRIDE datasets 2013–2025 (n = 36,249) with at least one top-level file of each type (file names; compression stripped; /generated/ paths and zip contents excluded). Periods: 2013–16 (n = 2,726) vs 2023–25 (n = 15,565). Standard results are submitter-deposited mzIdentML or mzTab files; repository-generated files are excluded. b–c, open spectra by PRIDE submission type (COMPLETE submissions must include peak lists). COMPLETE fell from 29% to 15% of datasets, accounting for 5–10 of the 13-point fall when changes in submission-type composition and within-type rates were calculated in both orders. Shading: 95% CI, group bootstrap. File presence does not establish validity or reuse.



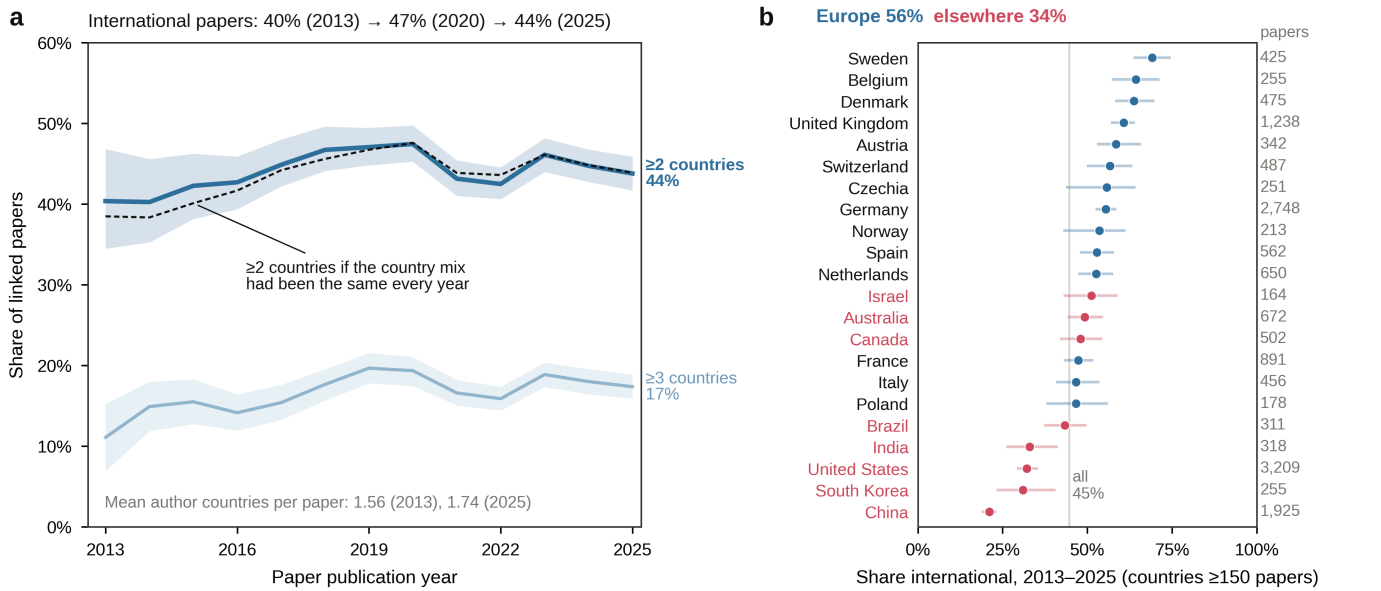
Supplementary Fig. 8 | Isobaric labelling and protein modifications. a–c, iTRAQ gave way to TMTpro, and the mean number of sample channels per labelled dataset grew from 6 to 13. PRIDE datasets with isobaric labelling in metadata, 2013–2025 (n = 6,116). a, Number of sample channels (plex) stated in text, among the 4,269 datasets that state one (highest TMT plex stated is used); iTRAQ 61% → 4% and TMTpro 0% → 48% from 2013–16 to 2023–25. b, Mean channels per dataset: counts stated in text vs reporter channels counted independently in table headers (channels used ≤ kit size; years with ≥20 datasets). c, isobaric datasets among all datasets. Shading: 95% CI, group bootstrap over contributor groups. **d–f,** PRIDE's post-translational modification (PTM) field names the modification in 81% of datasets with phosphorylation site tables and 60% with diglycine (GlyGly) site tables. PRIDE datasets 2013–2025 whose deposited MaxQuant output shows site tables (n = 6,113; the default Oxidation (M)Sites table makes them visible). A site table shows that the search allowed the modification as a variable modification; it does not show that samples were enriched for it. d, modifications with a site table. e, of datasets with that site table, share whose metadata names the modification. f, Share with a phosphorylation site table, by which metadata sources name phosphorylation. Whiskers: 95% CI, group bootstrap over contributor groups. Absence means "not detected". MaxQuant only.



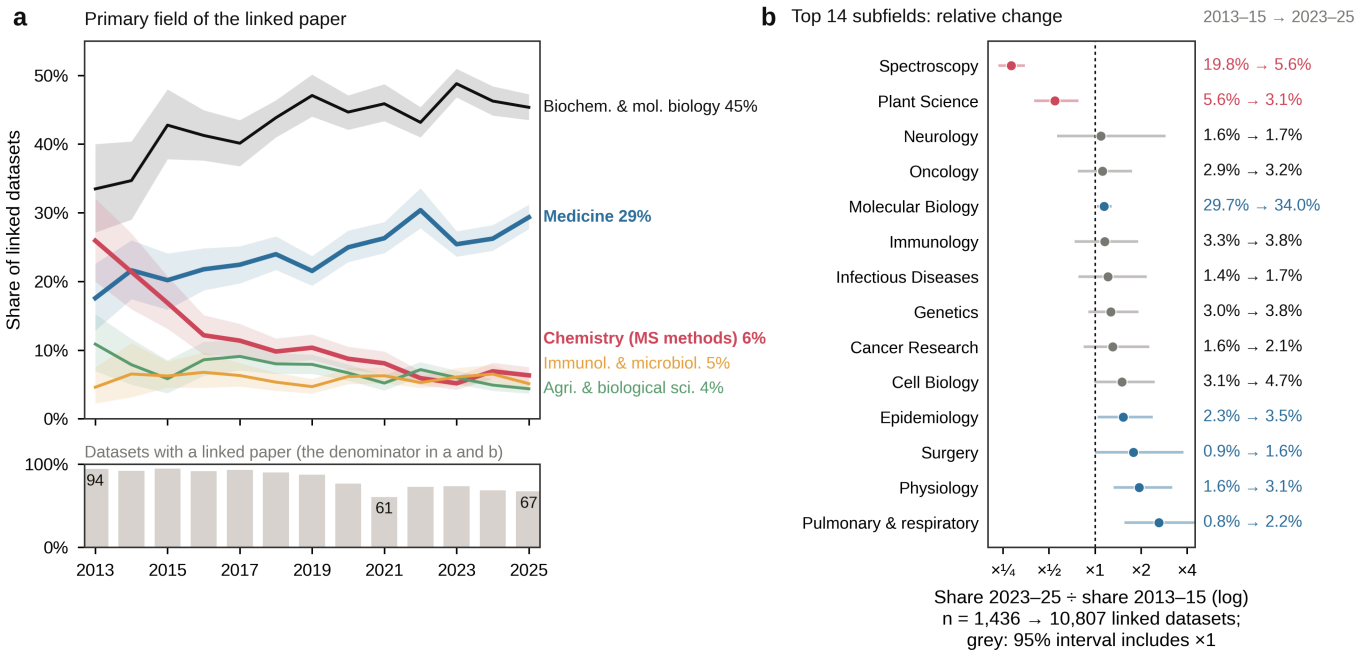
Supplementary Fig. 9 | Organisms. a–b, PRIDE deposits became more concentrated on human and mouse: datasets naming neither fell from 46% to 33%. PRIDE datasets 2013–2025 (n = 36,249; 8 list no species). Species from PRIDE records, with strains grouped under their species name; a dataset naming several species (2,931) counts once for each, so shares do not sum to 100%. Title and b: 2013–16 (n = 2,726) vs 2023–25 (n = 15,565). Within the 354 contributor groups depositing in both periods, the share naming neither human nor mouse also fell, 40% (34–46) → 32% (27–38). Shading and bars: 95% CI, group bootstrap. Dataset counts rose for every species shown; only their shares fell.



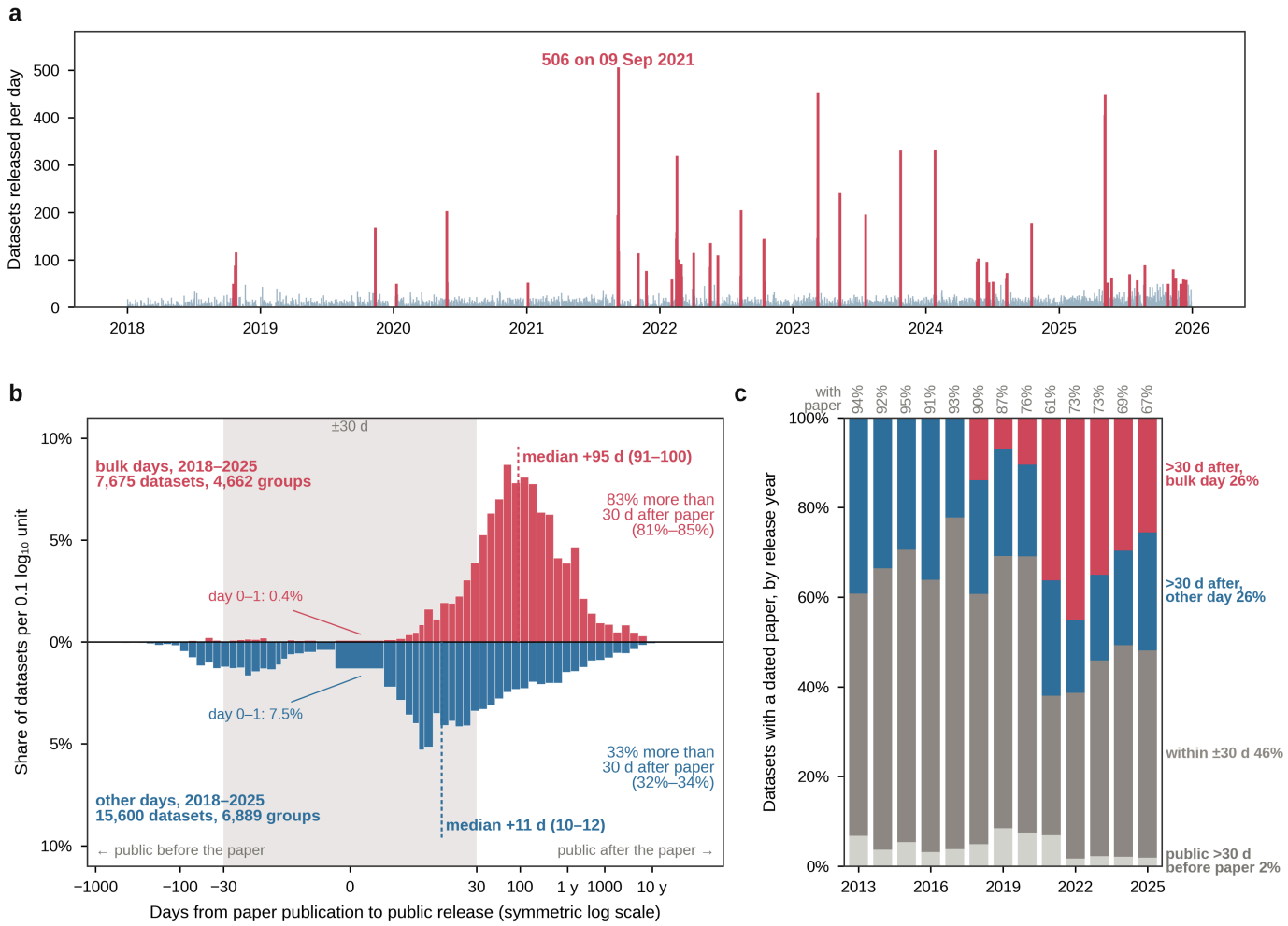
Supplementary Fig. 10 | Countries. a, China's share of public PRIDE datasets grew fivefold. Share of PRIDE datasets with an assigned country, by contributor country and release year (28,225 datasets). **b–d,** In the selected comparison, median data age at submission is 1.7–2.0 years for the UK, France and Spain, against 1.4 overall. Days from the latest acquisition date in raw-file names to PRIDE submission; only 11,868 of 35,364 datasets with raw files (34%) have dated names. (b) Thermo Fisher only, no detected data-independent acquisition (DIA), submitted 2016–2025, countries with ≥100 such datasets (9,109; the pooled median covers all of them). No country's median is clearly below the pooled median. Dots: median; bars: 95% interval over 2,000 resamples of contributor groups. A file-name date is supplied by the laboratory and may record naming rather than acquisition.



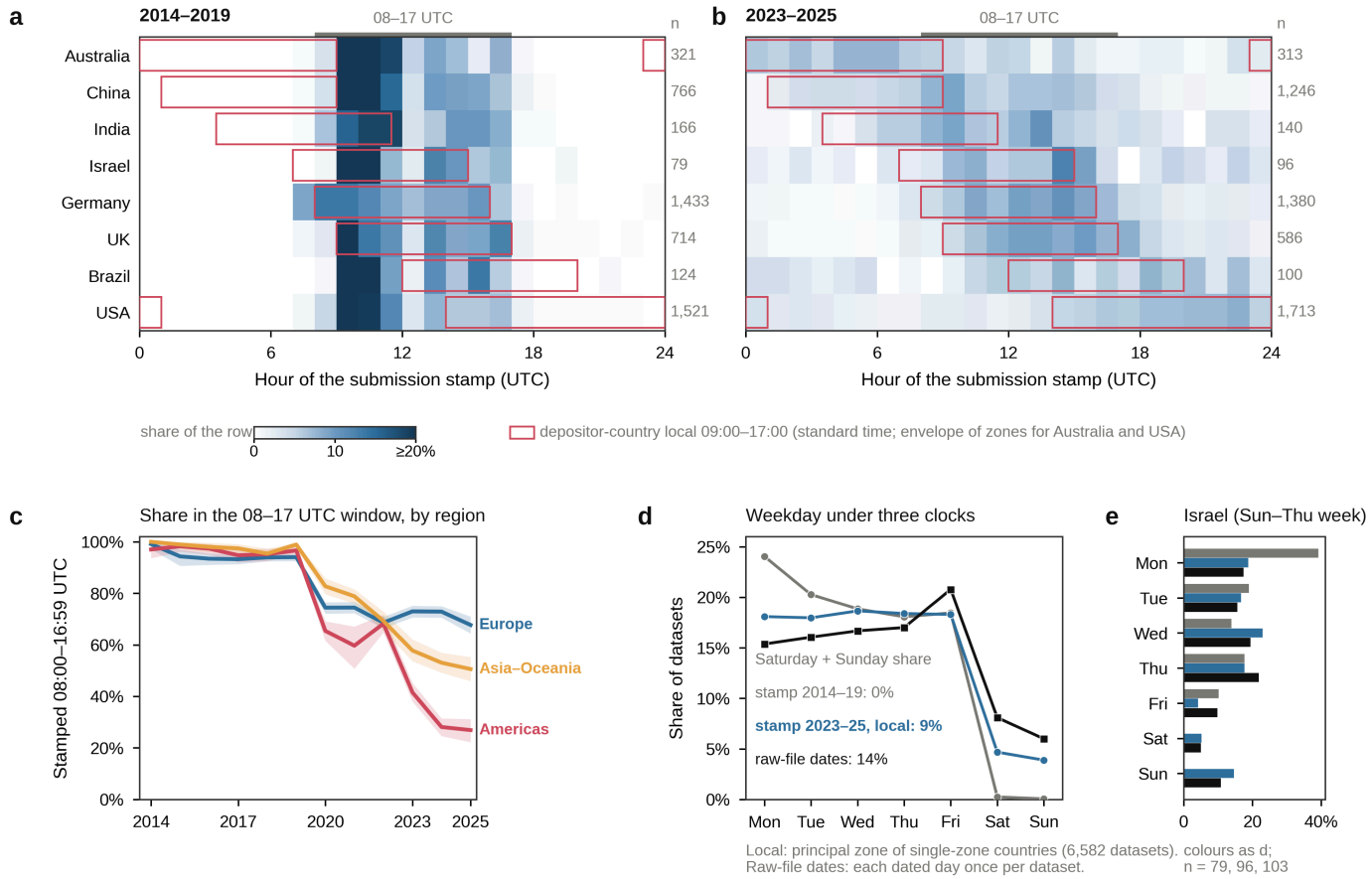
Supplementary Fig. 11 | Collaboration. a–b, International coauthorship occurs in 56% of European groups' papers, 32% of US groups' papers and 21% of Chinese groups' papers. Each paper is counted once: 22,939 OpenAlex works published 2013–2025 and linked as the primary paper of a PRIDE dataset (of 23,000; 61 lack author countries). International = authors' institutions in ≥2 countries. Country in b = the depositing group's country. Bands and bars: 95% bootstrap over groups.



Supplementary Fig. 12 | Research fields of linked papers. a–b, The share of datasets linked to mass-spectrometry methods papers fell from 26% to 6%; medicine rose from 18% to 29%. OpenAlex primary field/subfield of the paper linked to each PRIDE dataset, by release year: 27,031 of 36,249 datasets (2013–2025) have a linked paper. "Methods" = subfield Spectroscopy (95% of Chemistry). Each dataset is counted once; bands and bars: 95% bootstrap intervals over contributor groups.



Supplementary Fig. 13 | Release timing. a, Public releases cluster on a few dates: 25% of datasets since 2018 went public on 62 days. PRIDE datasets made public per day, 2018–2025. Red: days with ≥50 releases. Datasets released on those days went public a median 95 days after their paper (other days: 11). **b–c,** The share released >30 days after publication rose from 32% to 56% after 2020; the increase occurred on days with many releases. Public release minus publication date of the linked paper (OpenAlex): 27,001 of 36,209 datasets have a dated paper. Days with many releases: ≥50 PRIDE releases (62 days, all since Oct 2018). Release years 2013–20 vs 2021–25: >30 days after publication on other days 26% vs 22%; on days with many releases 7% vs 34%. Intervals: 95% over 2,000 resamples of contributor groups. Descriptive only: who set a release date is not recorded.



Supplementary Fig. 14 | Recorded submission times. a–e, Until 2019, 95% of PRIDE submissions were time-stamped 08–17 Coordinated Universal Time (UTC), wherever they came from. Submission timestamp of 35,946 of 36,249 datasets (the rest carry no time of day), as stored (UTC). From about 2020, recorded times follow depositors' working days (52% within 08–17 UTC in 2025). What the timestamp records is not documented; weekday patterns before 2020 describe this recorded clock rather than depositors' working patterns.